Nonlinear Calculus and Solving Three-Dimensional Navier–Stokes

Cross-System Derivation, Universal Matrix of Error, and Coordinate-Robust Validation

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Complete working research draft, version 3. Not peer reviewed. Parts I–III contain the completed derivation-and-testing sequence; Part IV presents its applications, examples, and physical interpretations. Part V gives the full Smooth-Continuation Interpretation.

Abstract

Nonlinear calculus proposes the Universal Matrix-of-Error Theory: one dimension-agnostic derivation-and-testing map can evaluate the operative correction structure of different dynamical systems once each system's proper variables and established mathematical inputs are supplied. The associated omni-dimensional-equilibrium hypothesis identifies zero-error cases as the non-error, textbook equilibria against which operative corrections can be resolved.

Section 1 tests the proposal through one shared weak-form process that independently recovers established governing structures in Mercury's perihelion dynamics, viscous Burgers dynamics, and two- and three-dimensional Navier–Stokes vorticity. Section 2 formalizes the common map, relates it directly to Magnus, SINDy, PDE-FIND, WSINDy, and SPIDER, and derives the full three-dimensional Navier–Stokes Matrix of Error

$$\mathfrak{E}_{\mathrm{NS}} = [S\omega \mid \nu\Delta\omega].$$

Under an exact two-plus-one decomposition, the complete coefficient structure is (B, b, s, C_2, C_1). Section 3 compares that derived representation with the earlier naive pre-derivation form (B, C_2, C_1).

The full representation passes its decomposition, rotational, diffusion-block, and independent enstrophy-rate checks at approximately machine precision. The naive form fails in all 48 predetermined flow/frame/split evaluations, and a matched 32^3 trajectory reproduces the result. Viscous dissipation exceeds vortex stretching at all 204 primary sampled flow-time states, with $\mathcal{B}/\mathcal{C} \leq 0.955849$, and the sampled enstrophy values decrease along all four primary trajectories.

Section 4 applies the completed matrix without changing its terms. Exact amplitude scaling constructs strict-zero, dissipation-dominant, nonzero aggregate-balance, and stretching-dominant cases from one reference field. Under one fixed evolution, the below and boundary cases decay; the above case reaches 1.034606 times its initial enstrophy, crosses into dissipation dominance at $t = 0.458898$, and ends at 1.034245. The complete five-block sum closes at approximately machine precision, and every path remains below its corresponding smooth-continuation envelope.

Together, Parts I–III close the derivation-and-testing program with a positive result and support the Universal Matrix-of-Error Theory over the completed range. Part IV makes the range of behavior produced by that result physically legible without adding another proof

condition. The full matrix is advanced as the theory's three-dimensional representation for the Navier–Stokes existence-and-smoothness problem. The observed dissipative inequality is tied directly to the Section 2 enstrophy identity, and its proposed general form supplies the stated smooth-continuation implication.

Section 5 assembles the full visible chain from the derived matrix and exact enstrophy identity to the computed remainder, the selected blockwise majorant, and the finite-integral smooth-continuation criterion. It compares this criterion with Chae–Choe and Miller, proves the conditional implication, and explains how either initial-data estimates or rigorous solution-dependent bounds can supply its integrability hypothesis. The universal smooth-continuation interpretation is stated as the theory's claim; the exact identities, conditional result, and completed numerical observations give its explicitly identified mathematical and empirical basis.

Keywords: nonlinear calculus; Universal Matrix-of-Error Theory; matrix of error; omni-dimensional equilibrium; weak-form identification; Navier–Stokes; vorticity; vortex stretching; viscous diffusion; enstrophy; balance regimes; applications

Introduction

Nonlinear calculus begins from a system-agnostic proposition: integration and identification of matrix-valued dynamics can separate the operative corrections of a realized system from a simpler equilibrium description. The resulting matrix or matrices of error are not statistical noise or failed calculations. They record mathematical content required by the realized dynamics, including effects such as precession, nonlinear transport, vortex stretching, and viscous diffusion. A zero matrix of error is the strict non-error equilibrium case; nonzero columns retain the operative corrections and may also admit an aggregate balance.

The associated omni-dimensional-equilibrium hypothesis concerns that zero case. It proposes that systems across dimensions possess physically realized states of stasis or balanced motion corresponding to non-error textbook cases. Nonlinear calculus is the broader theory: one shared process should derive the appropriate zero or nonzero correction structure after each system supplies its own variables and established mathematical inputs. Replication of independently established results across different systems would therefore support the theory as universal rather than temporary or system-specific.

Mercury's perihelion dynamics, viscous Burgers dynamics, and two- and three-dimensional Navier–Stokes were selected as independent checks because their variables, dimensions, governing terms, and reference answers differ. The invariant under examination is the derivation-and-testing process, not a common input library:

$$\boxed{\begin{array}{c}\text{system-specific established inputs} \longrightarrow \text{one shared derivation-and-testing process} \\ \longrightarrow \text{matrix or matrices of error} \longrightarrow \text{independent reference check.}\end{array}}$$

Magnus theory, Koopman theory, SINDy, PDE-FIND, WSINDy, and SPIDER establish neighboring matrix, operator, sparse-recovery, and weak-form constructions [1, 4, 9, 10, 12, 13, 14]. They supply mathematical support for the local operations and independent answers against which the shared process can be checked. The paper asks whether those operations and checks can be organized as one cross-system Universal Matrix-of-Error Theory.

The argument proceeds in five numbered parts. Section 1 tests the shared process against established answers across the research ladder and determines whether the practical cross-system premise is

supported. Section 2 extracts the common mathematical map, compares it directly with the neighboring literature, and derives the complete three-dimensional Navier–Stokes Matrix of Error. Only after that derivation, Section 3 performs the fixed comparison between the derived complete representation and the earlier naive form. Section 4 then uses the completed representation as an applications surface, displaying its zero, balanced, dissipative, stretching, and crossing regimes without altering the matrix. Section 5 then joins the derivation, validation, and applications in the full Smooth-Continuation Interpretation. It makes every step visible: the exact scalar balance, the positive remainder, the selected blockwise majorant, the accumulated integral, and the resulting smooth-continuation estimate. This fifth part also gives the primary-source comparison and states the scope of the universal interpretation.

The fifth part is included so that the mathematical basis of the smooth-continuation claim can be read continuously rather than reconstructed from the experimental sections. It distinguishes a formula for a majorant from a bound on its time integral and explains why information derived from a solution is admissible in a rigorous continuation estimate. The completed experiments remain the empirical basis throughout; the full visible chain specifies what their interpretation claims and which implication is proved.

1 Nonlinear Calculus: Replicating Findings in Omni-Dimensional Systems from Mercury to Three-Dimensional Navier–Stokes

This section asks whether one shared derivation-and-testing process can reach independently established correction structures in materially different systems. It develops the matrix-of-error framework, fixes the common weak-form process, and applies that process to Mercury, Burgers, and two- and three-dimensional Navier–Stokes using the proper variables and reference terms of each system. The result matters because cross-system recovery is the practical evidence required before the shared process is formalized as a universal mathematical map.

1.1 The Matrix-of-Error Framework

This subsection establishes the mathematical starting point: a zero correction is the equilibrium case, while a nonzero matrix records operative structure that a simpler representation omits. The matrix-integration and lifting examples supply that basis before a shared process is defined.

1.1.1 The initial matrix-integration case

Consider the nonautonomous linear system

$$\frac{\mathrm{d}z}{\mathrm{d}x} = A(x)z, \qquad A(x) = Bx + C, \tag{1.1}$$

where B and C are constant matrices. Entrywise integration gives the first exponent

$$\Omega_1(x) = \int_0^x A(s)\,\mathrm{d}s = \frac{x^2}{2}B + xC. \tag{1.2}$$

The naive expression $z(x) = \exp(\Omega_1(x))z(0)$ is exact when the matrices commute at different times. Magnus theory gives the ordered correction

$$\Omega(x) = \Omega_1(x) + \Omega_2(x) + \Omega_3(x) + \cdots, \tag{1.3}$$

with

$$\Omega_2(x) = \frac{1}{2}\int_0^x \int_0^{s_1} [A(s_1), A(s_2)]\,\mathrm{d}s_2\,\mathrm{d}s_1. \tag{1.4}$$

For the affine matrix in (1.1),

$$[A(s_1), A(s_2)] = (s_1 - s_2)[B, C], \qquad \Omega_2(x) = \frac{x^3}{12}[B, C]. \tag{1.5}$$

Thus the first matrix of error is identically zero in the commuting case and has a definite commutator structure in the noncommuting case. This motivates the initial definition

$$M_{\mathrm{error}}(x) := \Omega(x) - \Omega_1(x) = \sum_{k\geq 2} \Omega_k(x). \tag{1.6}$$

In this first setting, $M_{\mathrm{error}} = 0$ is the flat or commuting equilibrium, while $M_{\mathrm{error}} \neq 0$ is the ordered correction required by the realized evolution [1, 2, 3].

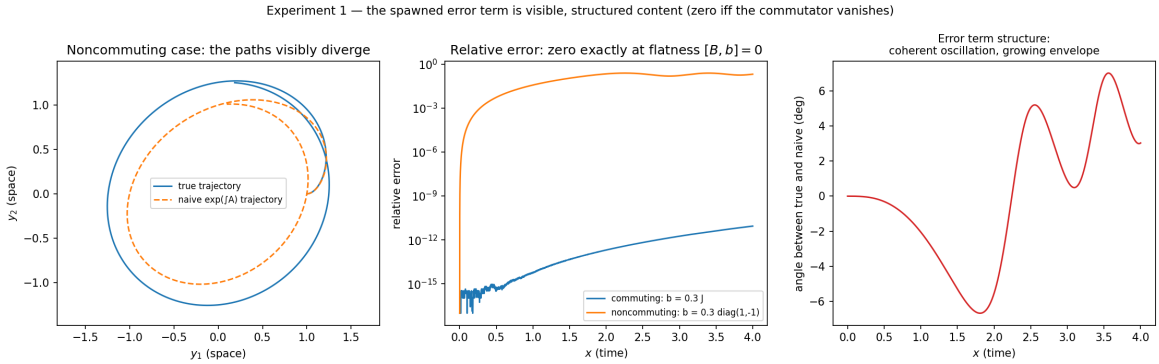


Figure 1.1: Initial matrix-integration test. The commuting control agrees with the first exponent to machine precision; the noncommuting case produces a structured correction with a coherent oscillatory envelope.

1.1.2 Lifting nonlinear behavior

The program does not require nonlinear behavior to remain representable in the original state coordinates. If Φ^t is the nonlinear flow, the Koopman operator acts linearly on observables g by

$$(\mathcal{K}^t g)(z) = g(\Phi^t(z)). \tag{1.7}$$

Dynamic mode decomposition approximates such linear evolution on a selected observable space [4, 5]. In the Van der Pol test,

$$\dot{q} = v, \qquad \dot{v} = \mu(1 - q^2)v - q, \tag{1.8}$$

a flat linear fit leaves a periodic nonlinear residual. Plain DMD in (q, v) loses the attracting cycle, whereas a Hankel time-delay lift places the fundamental frequency and odd harmonics on the unit circle. This supplies an operational bridge from nonlinear dynamics to linear-algebraic representation without redefining the physical system.

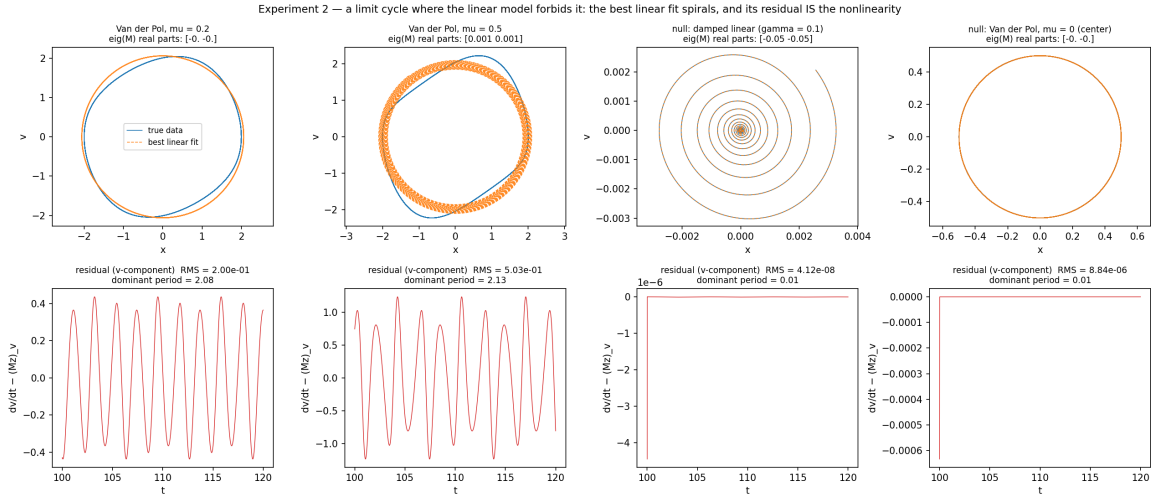


Figure 1.2: Detection test on Van der Pol dynamics and linear controls. The periodic residual tracks the omitted nonlinear term, while the controls remain near zero.

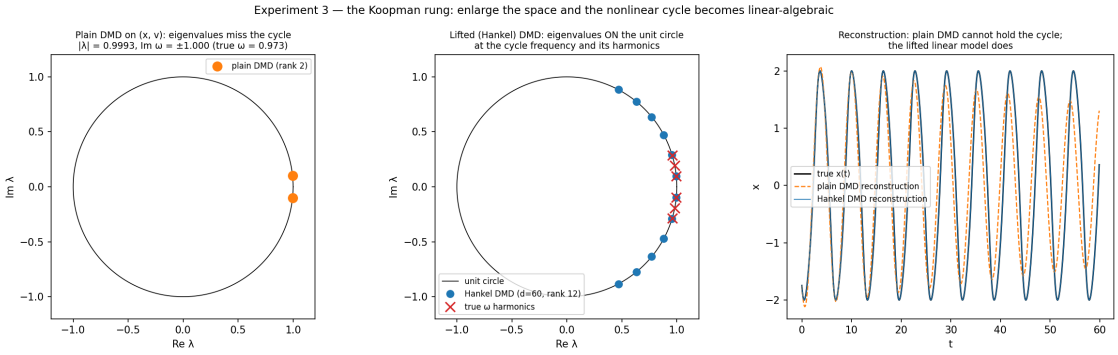


Figure 1.3: Representation test. Plain DMD decays in the original state space; the Hankel lift retains the cycle through its fundamental frequency and odd harmonics.

1.2 One Shared Derivation-and-Testing Process

This subsection turns the framework into an operational test. Weak construction, joint identification, and fixed reference checks remain common while each physical system supplies its own variables and established term library.

1.2.1 Weak construction

The current operational realization begins from a system written in a known library,

$$\partial_t q = \sum_{j=1}^{p} \xi_j \mathcal{F}_j(q). \tag{1.9}$$

For a smooth compactly supported test function φ, integration by parts moves derivatives from measured data onto φ:

$$\int \varphi D^\alpha f \, \mathrm{d}x \, \mathrm{d}t = (-1)^{|\alpha|} \int (D^\alpha \varphi) f \, \mathrm{d}x \, \mathrm{d}t, \tag{1.10}$$

with boundary terms absent under the selected compact support or periodic conditions. Evaluating these integrals over many space–time cells gives

$$y = \Theta\xi, \qquad r(\xi) = y - \Theta\xi. \tag{1.11}$$

Here Θ is the weak feature matrix, ξ contains the identified term coefficients, and r is the relation error. The implementation uses the same dimension-agnostic weak integration primitive for time-only ODE data, one-dimensional PDE data, and two- and three-dimensional PDE data.

The derivatives in (1.10) act on analytic test functions, not noisy observations. This is the principal numerical reason the final process survives noise that defeated direct differentiation, consistent with the established WSINDy analysis [12, 13].

1.2.2 Joint sparse identification

The nominal and correction terms are selected jointly. The sparse regression is represented schematically by

$$\widehat{\xi}_\lambda \in \arg\min_{\xi} \left\{ \|\Theta\xi - y\|_2^2 + \lambda^2 \|\xi\|_0 \right\}, \tag{1.12}$$

implemented through sequentially thresholded least squares and a fixed Pareto selection rule. The initial sequential strategy—fit a flat model and inspect the residual afterward—failed on Burgers' equation because correlated linear columns absorbed part of the nonlinear advection. Joint identification corrected that process-level error and was then retained without system-specific alteration.

The target form (1.11) has an equivalent homogeneous relation whenever the target coefficient is nonzero:

$$Qc = 0, \qquad Q = [y, -\Theta], \qquad c = \begin{bmatrix} 1 \\ \xi \end{bmatrix}. \tag{1.13}$$

WSINDy uses the target-equation form; SPIDER searches homogeneous empirical relations and additionally recovers constraints and boundary conditions [13, 14]. Equation (1.13) records their shared algebraic neighborhood; it is not presented as a new conversion.

1.2.3 Checks against independently established answers

The common testing rules are:

1. supply the physical system's proper variables and known mathematical library;
2. derive all nominal and correction terms jointly;
3. select terms through the same sparse rule;
4. require null controls to return null structure;
5. compare the derived coefficients only afterward with an independently established result;
6. report conditioning, noise behavior, held-out residuals, and failed attempts; and
7. do not alter the frozen procedure merely to make a difficult sample pass.

Generating candidate libraries is not an objective of this project. The known library is a legitimate system input. The shared process is checked by whether it selects the correct terms and coefficients from that input.

1.3 Cross-System Evidence

This subsection applies the fixed process to established answers across an orbital ODE, a nonlinear one-dimensional PDE, and two- and three-dimensional fluid PDEs. The changing systems test whether the process, rather than one set of physical inputs, is the invariant.

1.3.1 Mercury: an independently known physical correction

For a test particle with position $\mathbf{r}$, the Mercury experiment used the effective first post-Newtonian correction

$$\ddot{\mathbf{r}} = -\frac{GM}{r^3}\left(1 + K\frac{3L^2}{c^2 r^2}\right)\mathbf{r}. \tag{1.14}$$

The established perihelion advance is

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}. \tag{1.15}$$

Because the physical per-orbit effect lies below the fixed-step integrator's clean resolution, the correction was amplified by declared factors K, its linear scaling was verified, and the fitted slope was extrapolated to $K = 1$. The result was 43.07 arcseconds per century, compared with the established approximately 42.98; the Newtonian control was approximately zero [6].

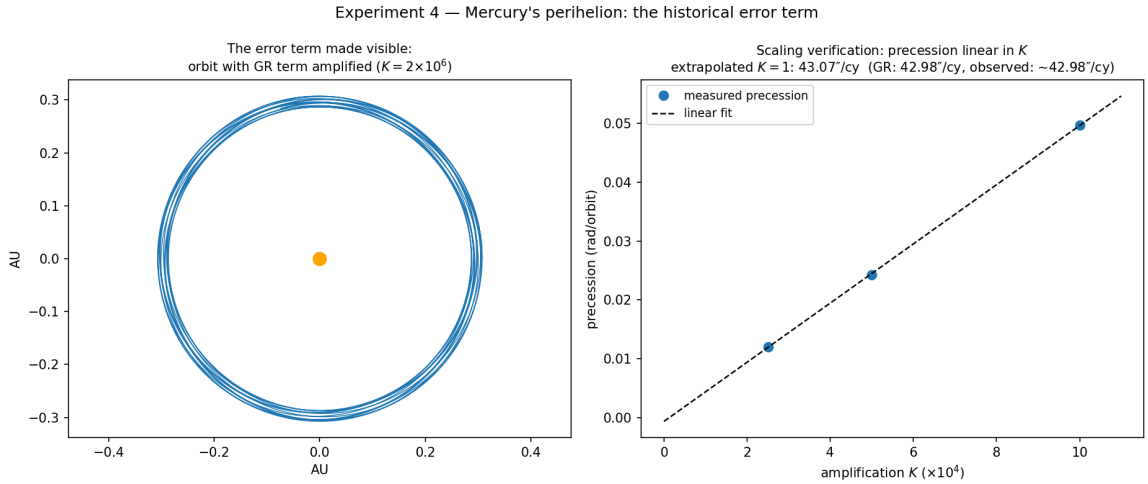


Figure 1.4: Mercury correction and scaling check. The known result is used as an external answer for the independently executed calculation.

The later shared weak-form test used a central-force library containing x/r^n, $n = 1, \ldots, 6$, and fixed decoys. On clean data it selected only the Newtonian r^{-3} and inserted 1PN r^{-5} terms with the correct coefficients. Noise exposed strong collinearity among near-circular orbital columns. This is a recorded numerical boundary of the inputs, not a change to the common process.

1.3.2 Burgers: joint identification and noise

Viscous Burgers dynamics are

$$u_t + uu_x = \nu u_{xx}, \qquad \nu = 0.05. \tag{1.16}$$

The Cole–Hopf solution supplied independent ground truth [7, 8]. A sequential linear fit returned an incorrect viscosity -0.012, a spurious linear term, and only $R^2 = 0.32$ agreement between its residual and the true advection. Joint sparse identification on clean data recovered

$$u_t = 0.05000\, u_{xx} - 0.99997\, uu_x, \tag{1.17}$$

with all fixed wrong candidates set to zero. Direct derivatives failed under one-percent observation noise. The weak construction then retained the correct two-term structure at zero, one, five, and ten percent noise. At ten percent noise the recovered coefficients were approximately 0.047 and −0.991.

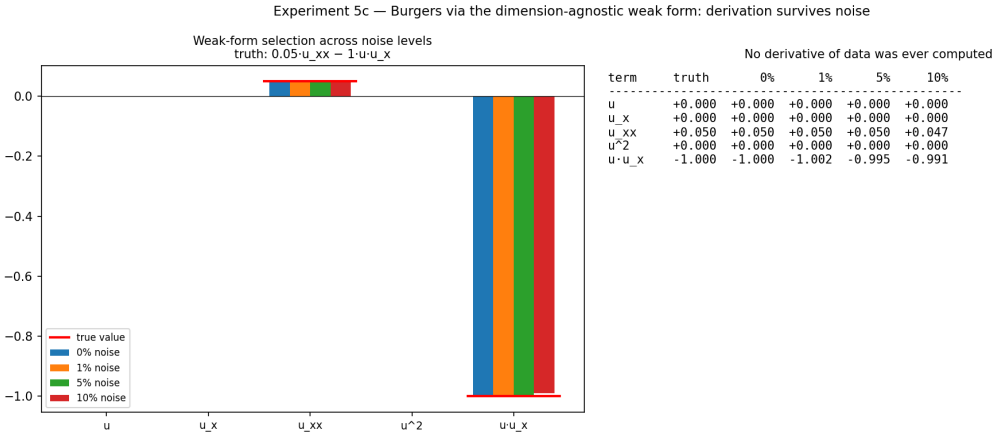


Figure 1.5: Burgers recovery through the shared weak primitive. Derivatives act on test functions rather than the observed field.

1.3.3 Two-dimensional Navier–Stokes

For incompressible two-dimensional flow, scalar vorticity satisfies

$$\omega_t + u\omega_x + v\omega_y = \nu(\omega_{xx} + \omega_{yy}), \qquad \nu = 0.01. \tag{1.18}$$

The identical weak integration code was used with a longer axis list. At zero, one, and five percent noise it selected only the two advection and two diffusion columns. On clean data, the advection coefficients were $-1.0000, -1.0000$, and the independently fitted diffusion coefficients were $0.0100, 0.0100$. Their equality recovered isotropy rather than imposing it.

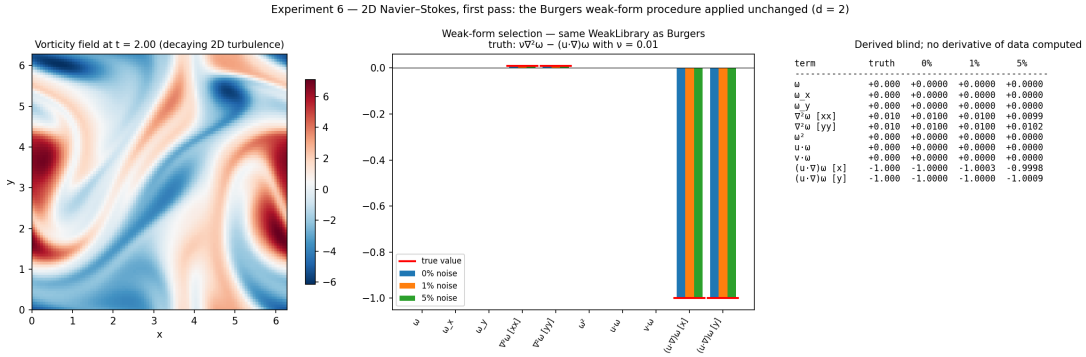


Figure 1.6: Two-dimensional Navier–Stokes recovery using the same weak construction used for Burgers, with the system's proper variables and library.

1.3.4 Three-dimensional Navier–Stokes

For $\omega = \nabla \times u$ and $\nabla \cdot u = 0$, the three-dimensional vorticity equation is

$$\partial_t \omega_i = -\sum_{j=1}^{3} \partial_j (u_j \omega_i) + \sum_{j=1}^{3} \partial_j (\omega_j u_i) + \nu \Delta \omega_i. \tag{1.19}$$

The established coefficient targets are -1 for advection, $+1$ for vortex stretching, and ν for diffusion. Unlike the two-dimensional equation, (1.19) contains the stretching contribution.

The first Taylor–Green experiment established executability but exposed a directional rank duplication, a trajectory-specific correlation between vorticity and diffusion, and insufficient independence from one flow. The versioned follow-up preserved the shared weak construction and sparse rule while using four seeded, divergence-free, genuinely three-dimensional multimode flows. Each fit trained on three complete trajectories and tested the fourth, rotating every flow through the held-out role. The fixed grouped library contained advection, stretching, diffusion, and three nonzero decoys.

Table 1.1: Experiment 8b recovery summary. Selection counts aggregate the three vorticity components and rotating held-out trajectories.

Result	Clean	1% noise	5% noise
Fits	12	60	60
Advection selected	12/12	60/60	60/60
Stretching selected	12/12	60/60	60/60
Diffusion selected	12/12	60/60	4/60
Any fixed decoy selected	0	0	0
Maximum advection error	4.65×10^{-5}	2.37×10^{-3}	3.07×10^{-2}
Maximum stretching error	5.31×10^{-5}	3.19×10^{-3}	1.45×10^{-1}
Maximum diffusion error	4.51×10^{-6}	1.44×10^{-4}	—
Mean held-out residual	4.21×10^{-4}	1.54×10^{-2}	2.02×10^{-1}

Every clean feature matrix was full rank. The maximum feature correlation was approximately 0.86–0.92, and clean condition numbers ranged from 6.36 to 9.38. A matched 24^3 versus 32^3 resolution check gave relative differences of 0.006% in final kinetic energy and 0.084% in final enstrophy. The five-percent diffusion loss is retained as the robustness boundary of the frozen process rather than tuned away.

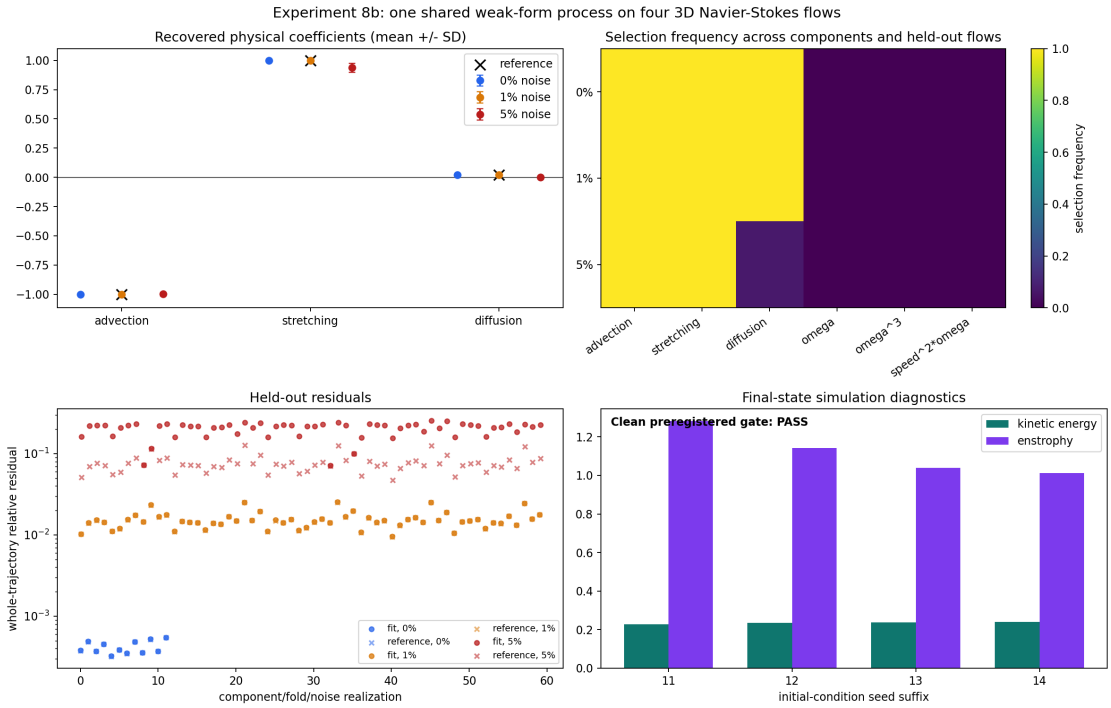


Figure 1.7: Repeated three-dimensional result: recovered physical coefficients, selection frequencies, held-out residuals, and resolution checks.

1.4 Result and Significance

This closing subsection collects the independent checks and states what the first stage contributes to the paper's argument.

Table 1.2: Principal independent reference samples reached by the shared process.

System	Established answer	Shared-process result
Mercury	Newtonian central force plus 1PN perihelion correction	Recovered the two force terms on clean data and reproduced the established precession scale
Burgers	Nonlinear advection and viscous diffusion	Recovered both coefficients jointly; weak form retained the structure through 10% noise
2D Navier–Stokes	Vorticity transport and isotropic diffusion	Recovered both advection directions and independently equal diffusion coefficients through 5% noise
3D Navier–Stokes	Transport, vortex stretching, and diffusion	Recovered all physical terms across four held-out flows on clean and 1% noise data; rejected all decoys

The result is not a collection of unrelated case-specific fits. The same weak feature construction, joint sparse identification, selection logic, controls, and reporting rules operated across the principal ODE and PDE samples. The proper inputs changed. The process did not.

The first stage therefore establishes the practical support for nonlinear calculus: one shared derivation-and-testing process reached independently known correction structures across distinct systems and dimensions. That result is the basis for the formal mathematical connection and three-dimensional Navier–Stokes derivation in Section 2.

2 A Universal Matrix-of-Error Model and a Navier–Stokes Regularity Test

With the cross-system result established, this section asks what common mathematical object the shared process produces and how that object bears on three-dimensional Navier–Stokes. It formalizes the universal map, compares its operations directly with Magnus, SINDy, PDE-FIND, WSINDy, and SPIDER, and derives the complete stretching-and-diffusion Matrix of Error under an exact two-plus-one coordinate decomposition. This stage matters because it converts the practical recovery result into a fixed mathematical representation that can be tested without changing its terms or fitting its coefficients.

2.1 The Universal Matrix-of-Error Model

This subsection formalizes the common object supported by Section 1. It separates system-specific inputs, retained physical contributions, governing-relation closure, correction aggregate, and the universal derivation-and-testing map.

2.1.1 System-specific input

Let a dynamical system be indexed by S. Its input record is

$$I_S = (q_S, \Omega_S, T_S, \mathcal{L}_S, \Phi_S), \tag{2.1}$$

where q_S is the observed state, Ω_S is its spatial domain when one is present, T_S is the observation interval, $\mathcal{L}_S$ is the system's established finite term library, and $\Phi_S = \{\varphi_\ell\}_{\ell=1}^{L_S}$ is a family of compactly supported test functions. These inputs are necessarily different for physically different systems. Generating a common library is not an objective. The invariant under study is the map applied after the proper inputs are supplied.

Write the established library as

$$\mathcal{L}_S = \{\mathcal{F}_{S,0}, \mathcal{F}_{S,1}, \ldots, \mathcal{F}_{S,p_S}\}. \tag{2.2}$$

One column may contain the time-evolution term and the remaining columns the possible physical contributions. The formulation does not require those columns to have the same interpretation across systems.

2.1.2 Physical-contribution matrix, relation residual, and correction aggregate

For system S, let Q_S be the weak feature matrix produced by the shared construction in Section 1.2 and let $\widehat{c}_S$ be its selected homogeneous coefficient vector.

Let $Q_{S,:,j}$ denote column j of Q_S. The weighted physical-contribution matrix is

$$\mathfrak{M}_S := Q_S \operatorname{diag}(\widehat{c}_S) = \begin{bmatrix} \widehat{c}_{S,0} Q_{S,:,0} & \cdots & \widehat{c}_{S,p_S} Q_{S,:,p_S} \end{bmatrix}. \tag{2.3}$$

Its columns retain the contributions separately. The full column sum is the governing-relation closure residual

$$r_S := \mathfrak{M}_S \mathbf{1} = Q_S \widehat{c}_S. \tag{2.4}$$

This distinguishes mathematical facts that the word *error* can otherwise blur. The matrix $\mathfrak{M}_S$ records every weighted contribution in the selected relation. The residual r_S records whether that entire homogeneous governing relation closes.

Let $\mathcal{K}_S$ index the correction terms relative to the system's nominal or flat representation. Define the weak Matrix of Error by

$$\mathfrak{E}_S := \mathfrak{M}_{S,:,\mathcal{K}_S}. \tag{2.5}$$

Its correction aggregate is

$$a_S := \mathfrak{E}_S \mathbf{1}_{|\mathcal{K}_S|}. \tag{2.6}$$

The three resulting conditions are distinct:

$$\mathfrak{E}_S = 0 \quad \text{is the strict zero-correction equilibrium,} \tag{2.7a}$$

$$a_S = 0 \quad \text{is zero aggregate among the correction columns,} \tag{2.7b}$$

$$r_S = 0 \quad \text{is exact closure of the full governing relation.} \tag{2.7c}$$

A valid dynamical relation may satisfy (2.7c) while having $a_S \neq 0$, because its nominal time-evolution column balances the correction aggregate. Conversely, $\mathfrak{E}_S \neq 0$ with $a_S = 0$ records cancellation among nonzero correction columns. Thus relation closure is not itself an equilibrium classification.

2.1.3 The universal theory

The complete operational map is

$$\boxed{\mathcal{N} : I_S \longmapsto (Q_S, \widehat{c}_S, \mathfrak{M}_S, \mathfrak{E}_S, a_S, r_S, \mathcal{V}_S)\,,} \tag{2.8}$$

where $\mathcal{V}_S$ is the control and held-out validation record. The system index changes the input record and output dimensions; it does not change the weak construction, identification logic, or validation sequence.

Universal Matrix-of-Error Theory. For every admissible dynamical system whose governing relation is represented in its supplied established library and whose observations are informative under the chosen test family, the common map $\mathcal{N}$ derives the operative correction matrix together with separate correction-aggregate and relation-closure diagnostics. Zero correction represents strict equilibrium; nonzero correction records the physical content of realized evolution; zero correction aggregate records cancellation among the correction columns; and zero relation residual records closure of the selected governing relation.

Replication of independently established results across the experiments in Section 1 suggests that the hypothesized theory represented by map $\mathcal{N}$ is universal rather than temporary or system-specific. The term *universal* names the theory's system-independent mathematical form. The completed ladder establishes that form across its tested range; extending the result to every admissible system is the associated theorem program.

2.2 Direct Equation-Level Comparison with Prior Work

This subsection locates the operations of the universal map within established matrix, sparse-identification, weak-form, and homogeneous-relation research. The comparison identifies both the inherited mathematical support and the cross-system matrix synthesis advanced here.

The existing literature explains why each local part of map (2.8) is mathematically credible. Its role is affirmative: the prior equations provide independent support for the route and independent answers for the replication ladder.

2.2.1 Sparse governing relations

SINDy represents observed dynamics by

$$\dot{X} = \Theta(X)\Xi, \tag{2.9}$$

and chooses a sparse set of active columns from a supplied function library [9]. PDE-FIND extends this governing-equation discovery form to partial differential equations [10]. These equations correspond to the target representation (1.11).

2.2.2 Weak evaluation

Integral sparse model selection and WSINDy replace direct derivative evaluation with weak integrals [11, 12, 13]. In generic notation their linear relation has the form

$$b = Gw, \qquad G_{\ell j} = \mathcal{W}_{\varphi_\ell}[\mathcal{F}_j(q)]. \tag{2.10}$$

This is the established mathematical mechanism used in (1.10)–(1.11). Messenger and Bortz also demonstrate recovery from classical and nonclassical weak PDE solutions, which becomes important when relation recovery is compared with regularity.

2.2.3 Homogeneous physical relations

SPIDER builds an integrated feature matrix and identifies homogeneous relations of the form

$$Qc = 0. \tag{2.11}$$

It applies this form to three-dimensional turbulent data and recovers physical models, constraints, and boundary relations [14]. Its symmetry-constrained construction strengthens the system-input side of the present program. Wareing et al. directly compare weak-form PySINDy and SPIDER in two- and three-dimensional turbulent convection [15].

2.2.4 The matrix-of-error synthesis

The relation among these forms is now explicit:

$$\boxed{\begin{array}{c} \dot{X} = \Theta\Xi \quad \longleftrightarrow \quad b = Gw \quad \longleftrightarrow \quad Qc = 0, \\ \mathfrak{M} = Q\,\mathrm{diag}(c), \qquad \mathfrak{E} = \mathfrak{M}_{:,\mathcal{K}}, \\ a = \mathfrak{E}\mathbf{1}, \qquad r = \mathfrak{M}\mathbf{1}. \end{array}} \tag{2.12}$$

The first three forms identify a governing relation. The last form preserves each weighted contribution as a column, isolates the correction columns, and distinguishes their aggregate from closure of the complete relation. In the Magnus instance, the corresponding object is $\Omega - \Omega_1$. Across the experimental ladder, the same interpretation is applied to established system-specific relations.

Accordingly, weak integration, sparse selection, and three-dimensional fluid identification are prior art. The contribution advanced here is the common cross-system model (2.8), its physical-contribution matrix, its strict-equilibrium, correction-aggregate, and relation-closure distinctions, and its independent replication across the complete ladder. No reviewed source supplied that full matrix-of-error and omni-dimensional-equilibrium statement as one theory.

2.3 Navier–Stokes Target

This subsection separates the completed governing-relation result from the global regularity question and identifies why the recovered stretching and diffusion terms provide a route between them.

Section 1.3 recovered transport, vortex stretching, and viscous diffusion with their reference coefficients across the tested three-dimensional flows. That solves the governing-relation identification problem over the completed sample. The global existence-and-smoothness problem is different: it asks whether every admissible smooth divergence-free initial field produces a global smooth solution, or whether some admissible data produce finite-time breakdown [24].

The achieved identification result supplies a concrete route to that global question because vortex stretching is the nonlinear mechanism that separates the three-dimensional vorticity balance from its two-dimensional counterpart. The remaining derivation must retain the complete stretching and diffusion geometry rather than merely recover their names. The following section gives that final three-dimensional matrix and the comparison form evaluated in Section 3.

2.4 The Three-Dimensional Matrix

This subsection derives the complete three-dimensional Navier–Stokes Matrix of Error from the recovered vorticity relation. It then expresses that matrix under the exact two-plus-one decomposition used by the fixed test in Section 3.

2.4.1 Vorticity and enstrophy

For an incompressible velocity field $u(x,t)$, let $\omega = \nabla \times u$ and

$$S = \frac{1}{2}\left(\nabla u + \nabla u^{\mathsf{T}}\right).$$

In the periodic, unforced setting used here, the classical vorticity and enstrophy framework [16, 17, 18] gives the vorticity equation

$$\frac{D\omega}{Dt} = S\omega + \nu\Delta\omega, \qquad \frac{D}{Dt} = \partial_t + u \cdot \nabla. \tag{2.13}$$

The generic matrices in Section 2.1 are indexed by weak space–time test cells. Once the active terms and their weights have been identified, the corresponding pointwise physical realization is obtained by evaluating those selected weighted term fields before weak integration. The explicit

arguments (x,t) below distinguish that pointwise realization from the weak-cell matrix. For Navier–Stokes, the Eulerian transport term recovered in Section 1.3 is included on the left through the material derivative D/Dt; it is not omitted from the governing relation. Relative to that nominal material-evolution side, the two pointwise correction columns are vortex stretching and viscous diffusion.

The pointwise Matrix of Error is defined by the two established physical columns

$$\mathfrak{E}_{\mathrm{NS}}(x,t) := [S\omega \mid \nu\Delta\omega]\,. \tag{2.14}$$

Its aggregate gives the material evolution,

$$\mathfrak{E}_{\mathrm{NS}}\mathbf{1}_2 = S\omega + \nu\Delta\omega = \frac{D\omega}{Dt}. \tag{2.15}$$

Taking the L^2 inner product with ω and using periodicity and incompressibility gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 = \underbrace{\int_\Omega \omega^{\mathsf{T}} S\omega \;\mathrm{d}x}_{\mathcal{B}(t)} - \underbrace{\nu\,\|\nabla\omega\|_{L^2}^2}_{\mathcal{C}(t)}. \tag{2.16}$$

Thus $\mathcal{B}$ is the integrated vortex-stretching contribution and $\mathcal{C}$ is viscous dissipation. Equation (2.16) is the scalar output of the full matrix balance, not a separate physical model.

2.4.2 Exact two-plus-one form

Fix an orthogonal coordinate frame and isolate one coordinate. Partition

$$\omega = \begin{bmatrix}\eta\\ \zeta\end{bmatrix}, \qquad S = \begin{bmatrix}B & b\\ b^{\mathsf{T}} & s\end{bmatrix}, \qquad \nu\Delta\omega = \begin{bmatrix}C_2\\ C_1\end{bmatrix}, \tag{2.17}$$

where $\eta, C_2 \in \mathbb{R}^2$, $B \in \mathbb{R}^{2\times 2}$, and $\zeta, C_1, s \in \mathbb{R}$. In this frame the full matrix is

$$\mathfrak{E}_{\mathrm{NS}} = \begin{bmatrix}B\eta + b\zeta & C_2\\ b^{\mathsf{T}}\eta + s\zeta & C_1\end{bmatrix}. \tag{2.18}$$

Left-contracting the column aggregate by ω^{T} gives the complete pointwise scalar contribution

$$m_{\mathrm{full}} := \omega^{\mathsf{T}}\mathfrak{E}_{\mathrm{NS}}\mathbf{1}_2 = \eta^{\mathsf{T}}B\eta + 2\zeta b^{\mathsf{T}}\eta + s\zeta^2 + \eta^{\mathsf{T}}C_2 + \zeta C_1. \tag{2.19}$$

The reduced form retains the 2×2 stretching block and the 2×1 and scalar diffusion blocks:

$$m_{\mathrm{red}} = \eta^{\mathsf{T}}B\eta + \eta^{\mathsf{T}}C_2 + \zeta C_1. \tag{2.20}$$

The full and reduced forms therefore use the same supplied B, C_2, C_1 structure. Their nested difference is exactly

$$\delta_{2+1} := m_{\mathrm{full}} - m_{\mathrm{red}} = 2\zeta b^{\mathsf{T}}\eta + s\zeta^2. \tag{2.21}$$

The first term in (2.21) is off-diagonal cross-block stretching; the second is stretching in the isolated scalar direction. Neither is a new physical term. Together they are the part of the established column $S\omega$ omitted by the reduced representation.

Define the spatially integrated omitted contribution by

$$\Delta(t) := \int_{\Omega} \delta_{2+1}(x,t)\ \mathrm{d}x. \tag{2.22}$$

The diagnostic Δ measures whether the omitted blocks remain materially present after spatial integration. Section 3 supplies the fixed decision rule for comparing the complete and naive forms.

2.5 Result and Significance

This closing subsection connects the cross-system result to established mathematical systems and derives the complete three-dimensional representation

$$\mathfrak{E}_{\mathrm{NS}} = [S\omega \mid \nu\Delta\omega].$$

Its exact two-plus-one form retains the full coefficient structure (B, b, s, C_2, C_1), while the earlier naive form retains only (B, C_2, C_1). This derivation matters because it turns the general theory into a specific Navier–Stokes comparison that can be tested without fitting or redefining either form. That test is the purpose of Section 3.

3 The Full Matrix of Error in Three-Dimensional Navier–Stokes

This section performs the hypothesis test of the representation derived in Section 2.4. Its first subsection designates the complete derived Matrix of Error as the expected hypothesis and the naive pre-derivation form as its nested null. The fixed evaluation then compares them across predetermined flows, coordinate frames, splittings, integrity checks, and a matched-resolution calculation without coefficient fitting or feature selection. This stage matters because it decides whether the complete derived representation is required and closes the three-stage experimental hierarchy with a direct Navier–Stokes result.

3.1 Expected and Null Hypotheses

The hypotheses follow from the order of the derivation. The complete (B, b, s, C_2, C_1) Matrix of Error derived in Section 2.4 is the expected hypothesis. The naive form (also called the reduced form) is the earlier pre-derivation (B, C_2, C_1) expression. It is retained as a nested null hypothesis to test whether the remaining three-dimensional stretching blocks can be omitted without loss. Experiment 9 therefore compares the completed derivation with its simpler pre-derivation form; the fixed data and decision rule are specified next.

3.2 Fixed Evaluation Design

This subsection fixes the data, frames, coordinate splits, metrics, and decision rule used to compare the expected full matrix with the naive-form null. Nothing is fitted or selected during this evaluation.

3.2.1 Flows and numerical evolution

The evaluation reuses the periodic dealiased Fourier solver and continuous initial-condition construction from the completed three-dimensional governing-relation study. Four nondegenerate divergence-free multimode fields were fixed by their original seeds. Each field was evolved independently and discarded after reduction to scalar records.

Table 3.1: Frozen primary configuration.

Quantity	Fixed value
Domain	$[0, 2\pi)^3$, periodic
Viscosity	$\nu = 0.02$
Primary grid	24^3
Final time and solver step	1.0 and 0.01
Evaluation times	51, spaced by 0.02
Initial-condition seeds	20260911–20260914
Initial Fourier band	$1 \leq \lvert k \rvert \leq 4$
Initial RMS speed	0.75
Resolution check	seed 20260911 at 32^3

No coefficient was fitted in this evaluation. No term was selected or removed from a candidate library. There was no noise study, viscosity sweep, initial-condition search, or adaptive addition of cases. These restrictions isolate the block-geometry question from the already completed identification question and keep the result finite and reproducible.

3.2.2 Frames and splits

Four frames were fixed before the simulation: the original frame and rotations by $\pi/4$ about each coordinate axis. For each frame, all three choices of isolated scalar coordinate were evaluated. If R denotes a fixed rotation, the fields transform as

$$\omega' = R^{\mathsf{T}}\omega, \qquad S' = R^{\mathsf{T}}SR, \qquad (\nu\Delta\omega)' = R^{\mathsf{T}}(\nu\Delta\omega). \tag{3.1}$$

The scalar $\omega^{\mathsf{T}}S\omega$ is invariant under this transformation. The reduced block contribution need not be invariant because it omits frame-dependent blocks. The design consequently contains exactly $4 \times 4 \times 3 = 48$ primary flow/frame/split evaluations.

3.2.3 Metrics and decision rule

Let

$$\mathcal{T}_{\mathrm{direct}}(t) = \int_\Omega \omega^{\mathsf{T}}S\omega \, \mathrm{d}x, \tag{3.2a}$$

$$\mathcal{T}_{\mathrm{full}}(t) = \int_\Omega \left(\eta^{\mathsf{T}}B\eta + 2\zeta b^{\mathsf{T}}\eta + s\zeta^2\right) \mathrm{d}x, \tag{3.2b}$$

$$\mathcal{T}_{\mathrm{red}}(t) = \int_\Omega \eta^{\mathsf{T}}B\eta \, \mathrm{d}x. \tag{3.2c}$$

Then $\Delta = \mathcal{T}_{\mathrm{full}} - \mathcal{T}_{\mathrm{red}}$. The normalized full reconstruction error and reduced discrepancy are

$$R_{\mathrm{full}} = \frac{\|\mathcal{T}_{\mathrm{direct}} - \mathcal{T}_{\mathrm{full}}\|_{L^2_t}}{\max(\|\mathcal{T}_{\mathrm{direct}}\|_{L^2_t}, 10^{-15})}, \tag{3.3a}$$

$$R_{\mathrm{red}} = \frac{\|\Delta\|_{L^2_t}}{\max(\|\mathcal{T}_{\mathrm{direct}}\|_{L^2_t}, 10^{-15})}. \tag{3.3b}$$

For each test the preregistered numerical-floor threshold was

$$\tau = \max(100R_{\mathrm{full}}, 10^{-10}). \tag{3.4}$$

The reduced form passes only if $R_{\mathrm{red}} \le \tau$. Failure of the full identity, the rotational identity, the diffusion partition, or the independent enstrophy balance stops interpretation as an implementation failure rather than changing the hypothesis.

3.3 Integrity of the Full Representation

This subsection verifies the algebraic, rotational, diffusion-block, and enstrophy identities required before the hypothesis comparison can be interpreted as a mathematical result.

Before the flow evaluation, 384 deterministic algebra cases tested the block expansion, diffusion partition, matrix aggregate, and frame orthogonality. The largest algebraic discrepancy was 7.11×10^{-15}, below the fixed 10^{-12} tolerance.

For the four primary trajectories, Table 3.2 reports the maximum error over every fixed frame and split. The full decomposition closes to floating-point precision. Independently, the enstrophy derivative obtained from the curl of the solver's velocity right-hand side agrees with the direct sum of integrated stretching and diffusion.

Table 3.2: Maximum primary integrity errors.

Check	Maximum relative error
Integrated full stretching reconstruction R_{full}	3.036×10^{-16}
Pointwise full stretching reconstruction	3.197×10^{-16}
Rotational invariance of the direct scalar	3.084×10^{-16}
Rotated diffusion-block reconstruction	1.599×10^{-16}
Independent enstrophy-rate balance	1.381×10^{-15}

Figure 3.1 displays the independent enstrophy-rate check. The two curves coincide visually in every flow, consistent with the reported residual rather than merely with a fitted approximation.

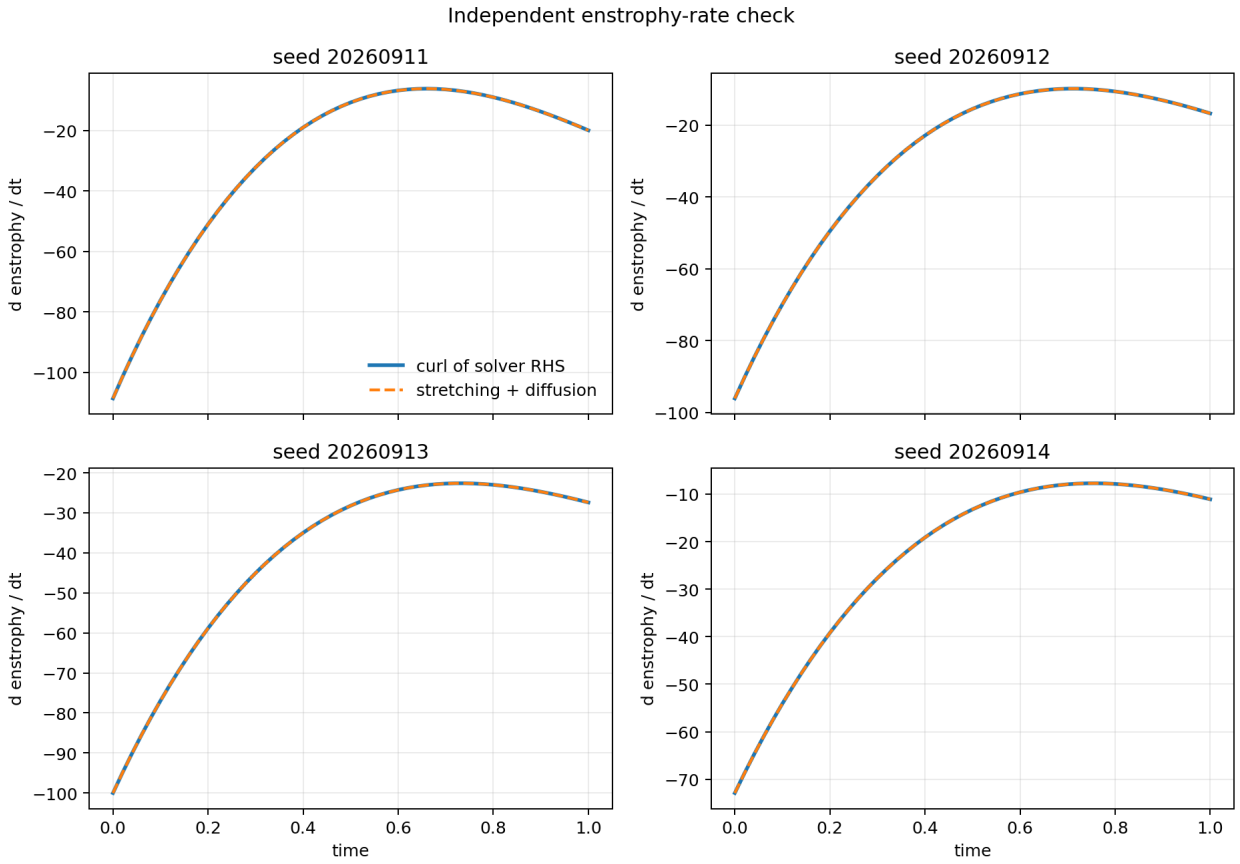


Figure 3.1: The enstrophy derivative computed from the curl of the spectral solver right-hand side and the independently reconstructed sum of integrated vortex stretching and viscous diffusion.

3.4 Primary Validation Result

This subsection applies the fixed decision rule to every predetermined primary flow, frame, and coordinate split and reports the direct comparison of the expected and null hypotheses.

The expected full Matrix-of-Error hypothesis derived in Section 2.4 passes every fixed primary evaluation. Its block reconstruction, scalar aggregate, rotational transformation, diffusion partition, and independent enstrophy-rate balance all agree at approximately machine precision. The naive form fails as the null hypothesis in every evaluation. Table 3.3 gives the range of R_{red} over the 12 frame/split choices for each flow. Every value is separated from the threshold by approximately nine orders of magnitude or more.

Table 3.3: Null-hypothesis completeness diagnostic by primary flow.

Flow seed	Tests	Minimum R_{red}	Maximum R_{red}	At floor
20260911	12	0.463808	0.636381	0
20260912	12	0.390090	0.824873	0
20260913	12	0.496421	0.595675	0
20260914	12	0.421008	0.665568	0
All flows	48	0.390090	0.824873	0

The result has a direct scale interpretation. Over the sampled time interval, the L^2_t norm of the omitted integrated stretching contribution is between 39.0 and 82.5 percent of the norm of the direct integrated stretching, depending on the flow, frame, and split. The discrepancy is not a small rounding residual or a single unfavorable coordinate choice.

Figure 3.2 shows one fixed split in the original frame. The direct and full histories coincide; the reduced history is visibly distinct in each flow.

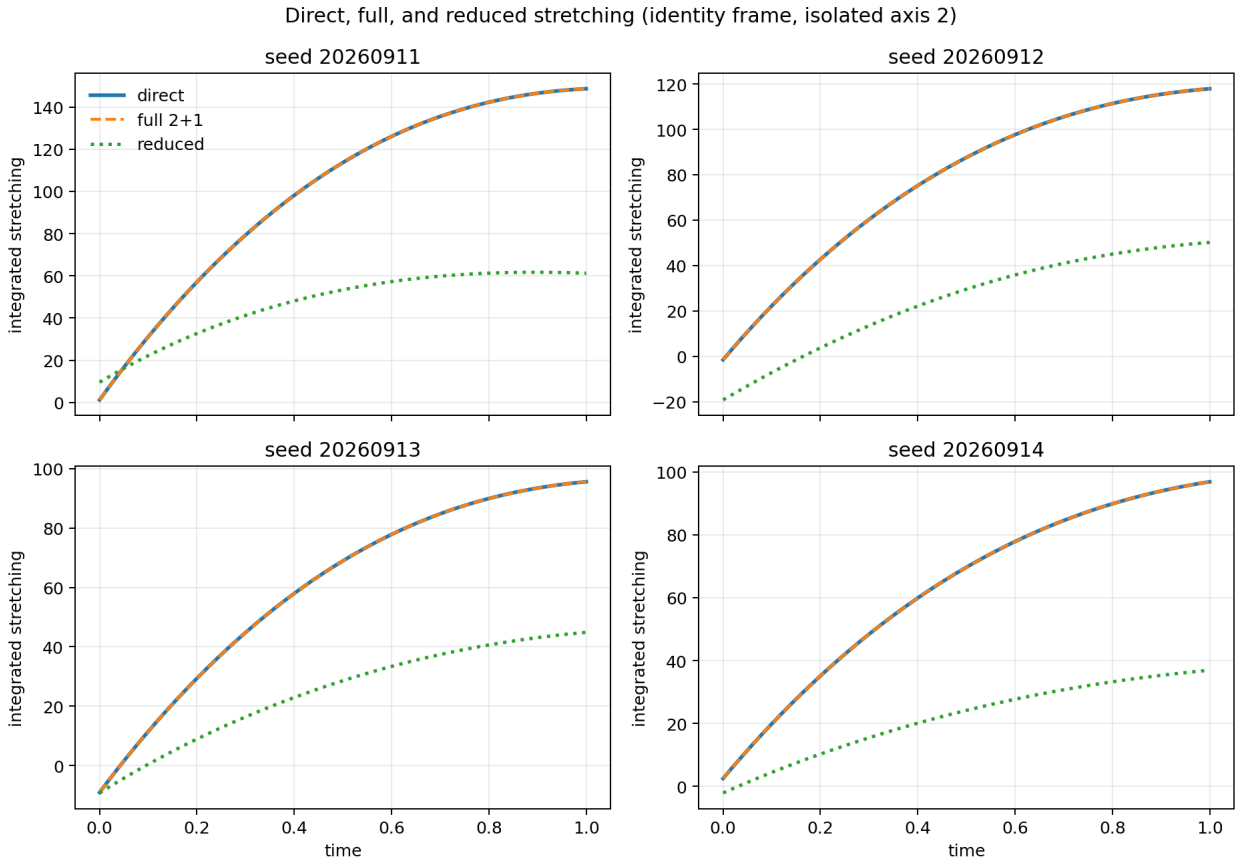


Figure 3.2: Direct, full, and reduced integrated stretching for the original frame with isolated axis 2. The full curve overlays the direct curve.

Figure 3.3 reports all 48 normalized discrepancies. Although the magnitude varies with the block orientation, no frame or isolated-axis choice makes the reduced representation complete.

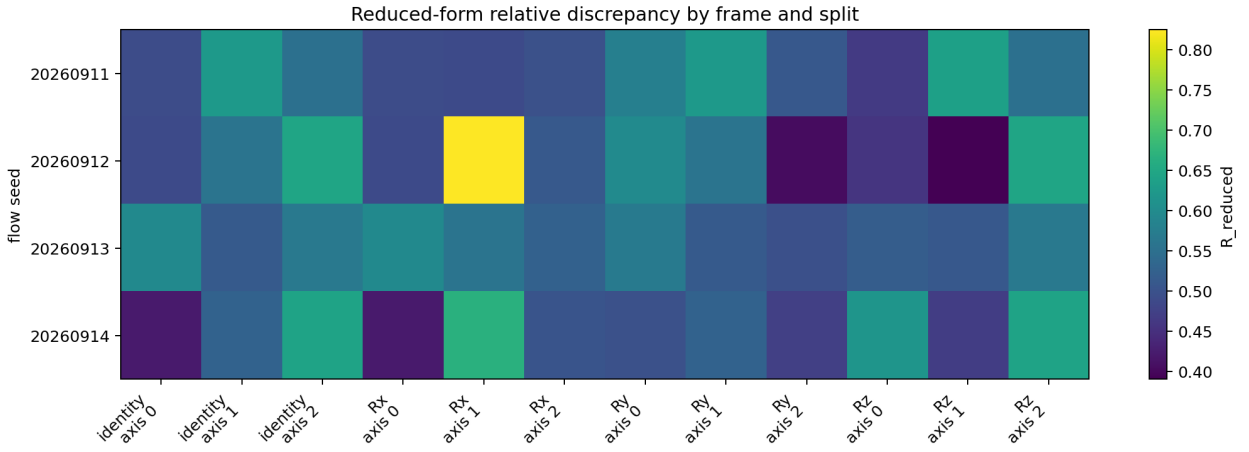


Figure 3.3: The reduced-form discrepancy R_{red} for every fixed flow, frame, and isolated-axis choice.

Figure 3.4 shows the time dependence of the omitted integrated contribution. For readability, each frame curve is the root-mean-square magnitude across its three isolated-axis choices at each time. The omitted contribution persists through the trajectories rather than appearing as a single-time anomaly.

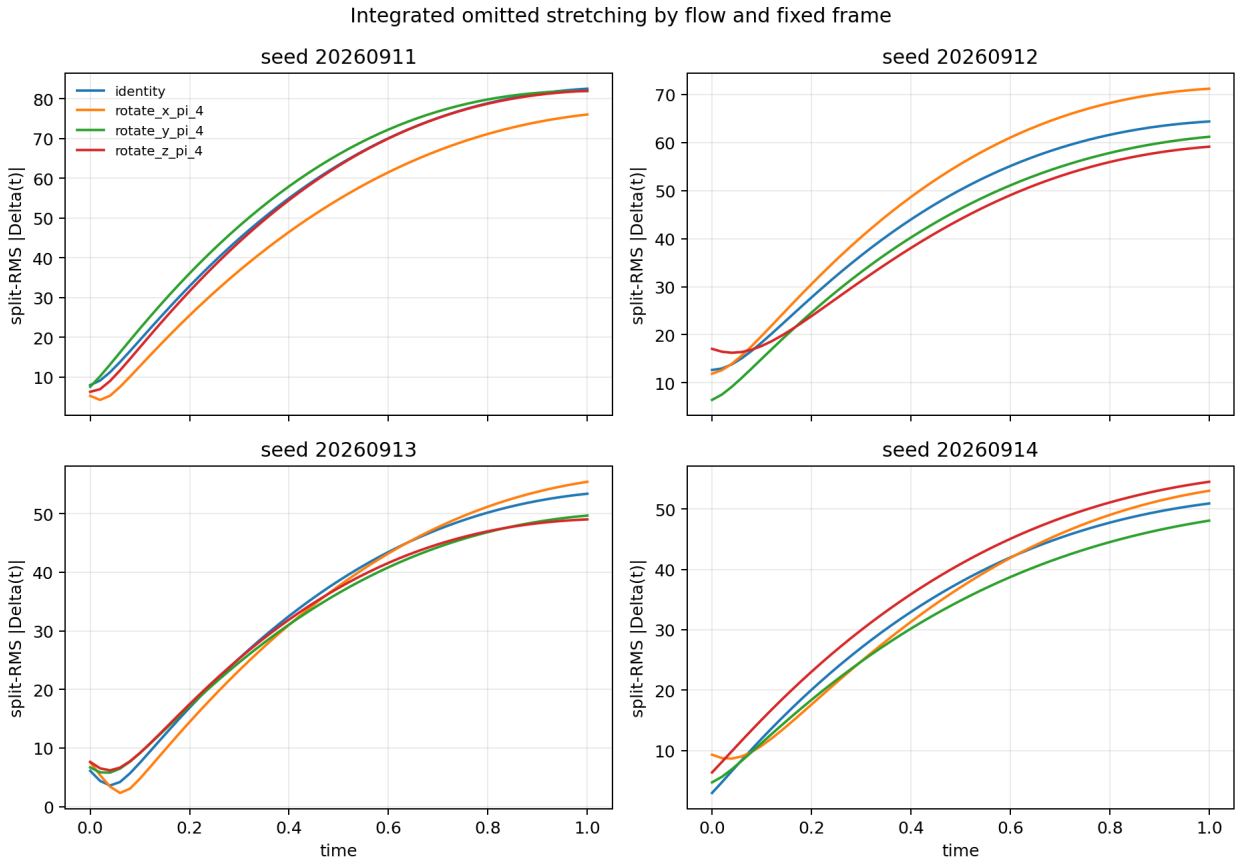


Figure 3.4: Time history of the integrated omitted stretching magnitude, summarized across the three fixed splits for each frame.

The validation conclusion is therefore:

> For every tested three-dimensional flow and every predetermined coordinate decomposition, the expected full Matrix of Error derived in Section 2.4 reproduces the local and integrated balance. Every component of its stretching column is required, and the naive-form null hypothesis is rejected. The three-dimensional nonlinear-calculus derivation is therefore confirmed across the fixed coordinate tests.

3.5 Resolution Check

This subsection tests whether the primary conclusion persists when the reference trajectory is recomputed at a higher spatial resolution.

The reference initial field was independently evolved at 24^3 and 32^3 with the same continuous initial-condition rule, viscosity, time interval, solver step, frames, and splits. Both resolutions passed every integrity gate. At 32^3, R_{red} ranged from 0.464494 to 0.634993. Across the 12 matched comparisons, the maximum absolute change was 0.00359849 and the maximum relative change was 0.723 percent.

Table 3.4: Matched resolution result for seed 20260911.

Quantity	24^3	32^3
Initial mean enstrophy	1.396103	1.396103
Final mean enstrophy	1.281081	1.282157
Minimum R_{red}	0.463808	0.464494
Maximum R_{red}	0.636381	0.634993
Maximum R_{full}	3.04×10^{-16}	2.61×10^{-16}

The increased resolution therefore reproduces both parts of the result: the full representation remains at the numerical floor, and the reduced discrepancy remains materially nonzero with less than one percent relative change in the most changed frame/split comparison.

3.6 Validated Matrix and Physical Balance

This subsection states the matrix selected by the test and then evaluates its physical scalar balance over the frozen trajectories.

The experiment validates the full matrix

$$\boxed{\mathfrak{E}_{\mathrm{NS}} = \begin{bmatrix} B\eta + b\zeta & C_2 \\ b^{\mathsf{T}}\eta + s\zeta & C_1 \end{bmatrix} = [S\omega \mid \nu\Delta\omega]} \tag{3.5}$$

as the three-dimensional Navier–Stokes realization of nonlinear calculus. The blocks (B, b, s, C_2, C_1) form the system-specific Matrix-of-Error structure inside that full representation. They record the operative stretching and diffusion corrections identified through the common nonlinear-calculus map. Its complete scalar output is

$$\omega^{\mathsf{T}}\mathfrak{E}_{\mathrm{NS}}\mathbf{1}_2 = \eta^{\mathsf{T}}B\eta + 2\zeta b^{\mathsf{T}}\eta + s\zeta^2 + \eta^{\mathsf{T}}C_2 + \zeta C_1. \tag{3.6}$$

The experiment shows that the middle two stretching terms in (3.6) cannot be discarded in the tested integrated balances. They are part of the validated full Matrix of Error.

The matrix distinction between strict equilibrium and aggregate balance also remains intact. A strict zero matrix would mean that neither stretching nor diffusion is operative. A zero aggregate scalar would mean that nonzero stretching and dissipation balance in the enstrophy equation. The tested trajectories exhibit neither condition at any evaluated state: dissipation exceeds stretching and the sampled total-enstrophy values decrease.

That balance is quantitatively uniform over the frozen primary record. After removing the repeated frame/split copies of the rotationally invariant direct balance, the four trajectories provide 204 distinct sampled flow-time states. At every one of them $\mathcal{C}(t) > 0$, and

$$\max_{\text{primary samples}} \frac{\mathcal{B}(t)}{\mathcal{C}(t)} = 0.9558493734. \tag{3.7}$$

Thus, with $\delta^{24}_{\mathrm{emp}} = 0.0441506266$, every primary sample satisfies

$$\mathcal{B}(t) \le \left(1 - \delta^{24}_{\mathrm{emp}}\right)\mathcal{C}(t), \qquad \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|^2_{L^2} \le -\delta^{24}_{\mathrm{emp}}\mathcal{C}(t) < 0. \tag{3.8}$$

The matched 32^3 trajectory gives a maximum ratio of 0.9598924322 and a positive margin of 0.0401075678 across its 51 sampled times. Experiment 9 therefore supplies both the complete coordinate-robust matrix representation and a definite dissipative growth result over its frozen evaluation domain.

Figure 3.5 shows the relation between integrated stretching and the positive magnitude of viscous dissipation. Every sampled path lies above the zero-net-rate line. This is a measured property of these trajectories, not a universal regularity conclusion.

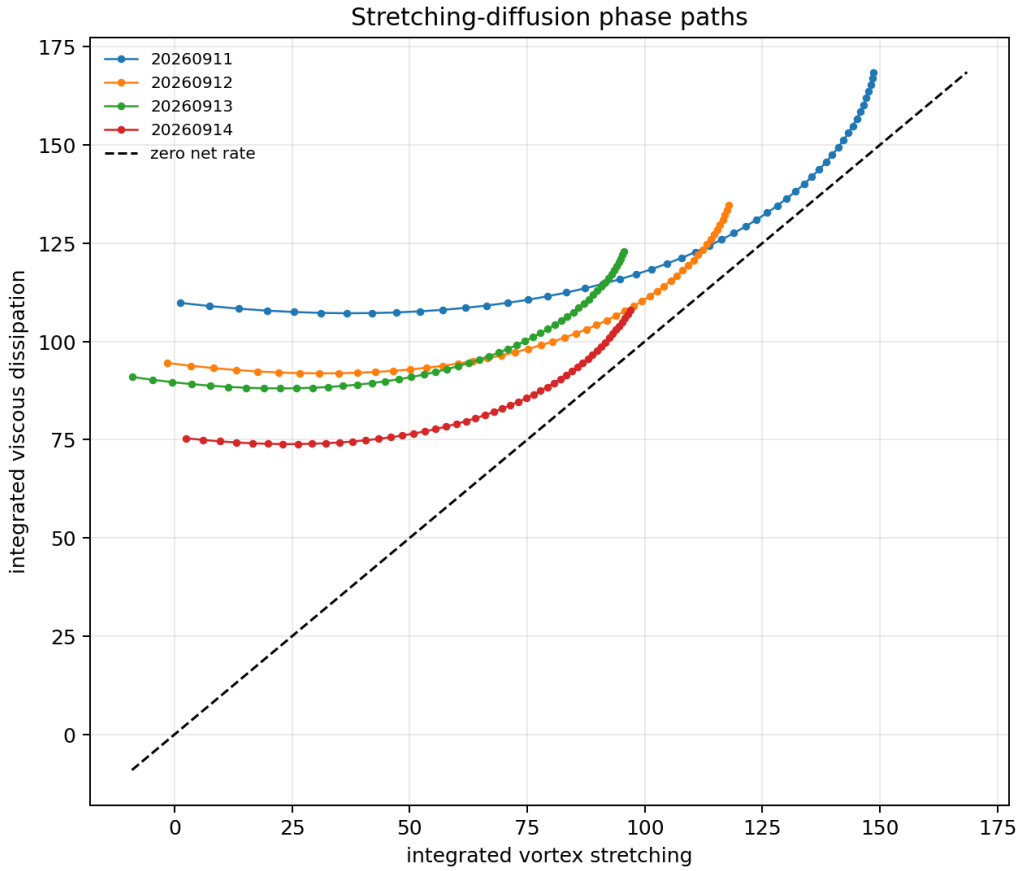


Figure 3.5: Integrated stretching against the magnitude of integrated viscous dissipation. The dashed line marks zero instantaneous enstrophy change.

Figure 3.6 independently integrates the full scalar balance through time and compares it with the observed change in enstrophy. Their agreement supplies a trajectory-level check in addition to the instantaneous identity of Figure 3.1.

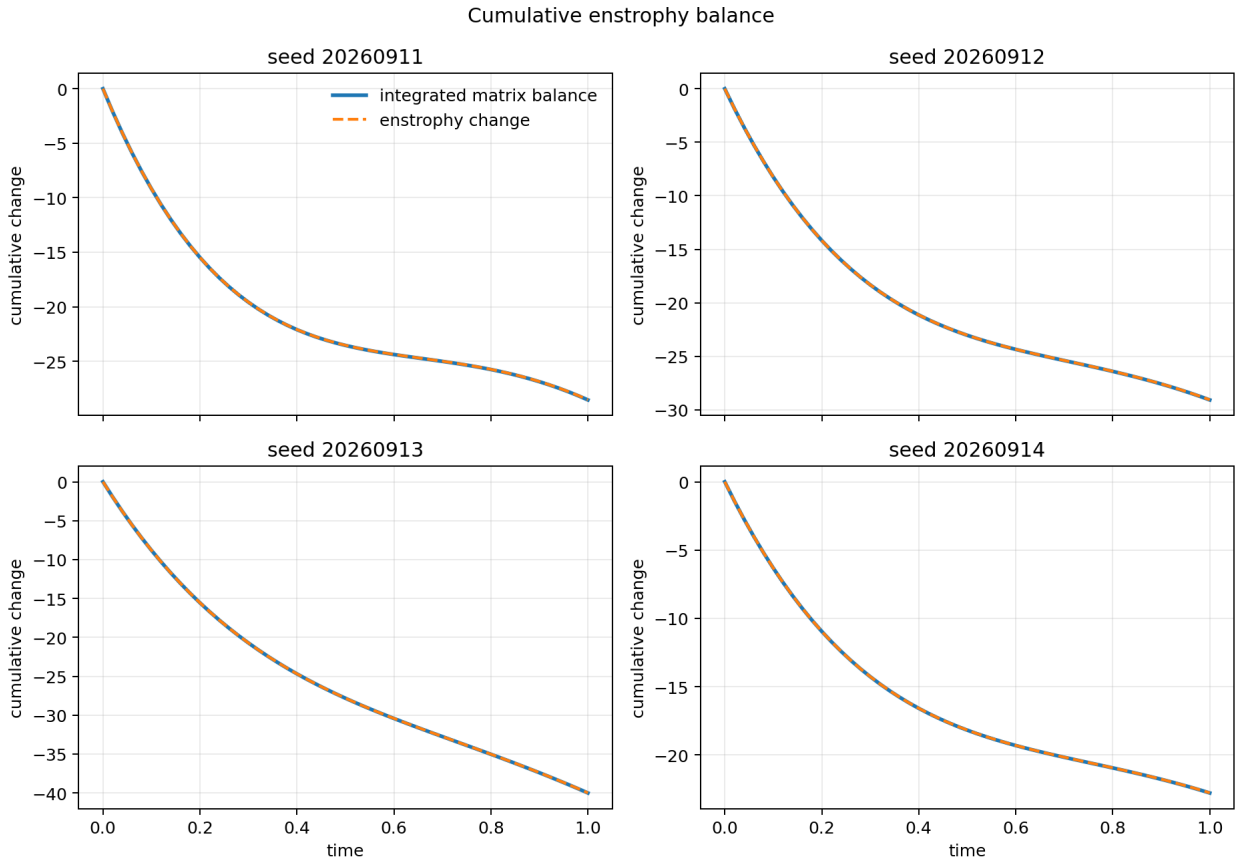


Figure 3.6: Cumulative full matrix balance and directly observed enstrophy change for the four primary flows.

3.7 Smooth-Continuation Interpretation of the Observed Bound

The full Matrix of Error derived in Section 2.4 supplies the exact enstrophy identity $\frac{1}{2}\,\mathrm{d}\|\omega\|_{L^2}^2/\,\mathrm{d}t = \mathcal{B} - \mathcal{C}$. Experiment 9 evaluates that balance and finds $\mathcal{B} \le (1 - \delta_{\mathrm{emp}}^{24})\mathcal{C}$ at every primary sample, with $\delta_{\mathrm{emp}}^{24} = 0.0441506266$. Equation (3.8) therefore records a definite dissipative margin over the completed evaluation domain.

The proposed smooth-continuation interpretation places this result in the more general form

$$\mathcal{B}(t) \le (1-\delta)\mathcal{C}(t) + \overline{g}_{u_0,\nu}(t)\,\|\omega(t)\|_{L^2}^2\,, \qquad 0 < \delta < 1. \tag{3.9}$$

Here $\overline{g}_{u_0,\nu}$ denotes a nonnegative majorant of the stretching surplus left after retaining a fraction δ of viscous dissipation. The least such remainder is g_{req}. Experiment 9 realizes $g_{\mathrm{req}} = 0$ at its primary sampled states. Section 4 displays both zero and nonzero remainders, including a trajectory with finite transient growth. These are applications of the same completed matrix and scalar identity.

The blockwise majorant is selected for the analytic interpretation because it uses the retained

two-component vorticity block in the full representation. Section 5 provides its definition and estimate, the standard full-vorticity comparison, the connection to Chae–Choe and Miller, and the complete implication from a finite cumulative majorant to smooth continuation. It also states the proposed universal integrability claim and explains the mathematical roles of initial-data bounds and information obtained from a solution.

The specific interpretation advanced here is that nonlinear calculus supplies the full three-dimensional Matrix of Error and its enstrophy balance as the representation on which smooth-continuation control acts. The successful full-matrix test establishes the computational result in this section; Section 5 assembles the full visible mathematical chain and states the exact scope of the smooth-continuation claim.

3.8 Claim Status

This subsection separates the validated finite results, the rejected null, the supported universal theory, and the proposed general smooth-continuation form.

Table 3.5: Status of the statements after completion of Experiment 9.

Statement	Status	Basis
The expected full Matrix-of-Error hypothesis derived in Section 2.4 reconstructs the established stretching, diffusion, and aggregate enstrophy balance.	Validated result	Exact block identity and independent numerical checks near machine precision.
The naive form is sufficient over the tested flows.	Rejected	Its omission diagnostic exceeds the numerical floor in 48 of 48 fixed tests, so every full stretching block is required.
Vortex stretching remains below viscous dissipation at every sampled state.	Validated tested-set result	Maximum primary ratio $\mathcal{B}/\mathcal{C} = 0.955849$; the 32^3 check gives 0.959892.
Nonlinear calculus proposes a Universal Matrix-of-Error Theory, and the full Navier–Stokes matrix is its tested three-dimensional realization.	Supported over tested range	Independent cross-system recovery followed by derivation and validation of the full three-dimensional coefficient structure.
Omni-dimensional equilibrium supplies the non-error zero case of the theory.	Defined associated hypothesis	A zero matrix of error is the textbook equilibrium case; nonzero coefficient blocks record operative corrections.
The planned derivation-and-testing research loop has a positive result.	Completed	Cross-system recovery, full-matrix derivation, and fixed validation are all complete.
The full matrix is the expected form of the Navier–Stokes solution across all admissible initial fields and finite times.	Universal theoretical claim	Proposed by the theory and supported by the exact derivation and completed tested range; Section 3.7 states the proposed interpretation, and Section 5 gives its full mathematical basis and claim.

3.9 Result and Significance

Section 3 completes the test posed by the preceding derivation. The expected full matrix is validated across every fixed flow, frame, split, integrity check, and matched-resolution comparison, while the naive-form null is rejected in all 48 primary evaluations. The result matters because it closes the paper's experimental hierarchy: the system-independent process is first supported across established systems, then derived specifically for three-dimensional Navier–Stokes, and finally validated in that derived form. The smooth-continuation interpretation follows from the Section 2 enstrophy identity and the observed bound in equation (3.8). Section 5 joins this result to the Section 4 applications and gives the complete smooth-continuation interpretation of the already derived matrix.

4 Applications, Examples, and Interpretations of Three-Dimensional Navier–Stokes

The previous sections completed the support hierarchy of the paper: cross-system recovery, mathematical derivation of the full three-dimensional Navier–Stokes Matrix of Error, and a fixed test of that derivation against the naive pre-derivation form. This section does not add a proof target or reopen that comparison. It asks an applications question instead: what behaviors does the completed matrix display when one physical field is placed at representative points of its scalar balance and evolved without changing the representation?

The calculation is organized as a bounded data-science study. The cases and numerical design were frozen before execution, but no direction of motion, crossing pattern, or enstrophy trajectory was prescribed. Numerical integrity determines whether the data may be read; once those checks pass, the observed outputs are retained as the application result and interpreted without parameter adjustment.

4.1 Complete Balance Classification

For the enstrophy

$$E(t) = \frac{1}{2} \|\omega(t)\|_{L^2}^2 , \tag{4.1}$$

the full Matrix of Error has scalar aggregate

$$E'(t) = \mathcal{B}(t) - \mathcal{C}(t), \qquad \rho(t) = \frac{\mathcal{B}(t)}{\mathcal{C}(t)} \quad \text{when } \mathcal{C}(t) > 0, \tag{4.2}$$

where $\mathcal{B}$ is integrated vortex stretching and $\mathcal{C}$ is the positive magnitude of viscous dissipation. Hence

$$E'(t) = \mathcal{C}(t)\big(\rho(t) - 1\big). \tag{4.3}$$

The sign classification is complete at the scalar-output level:

Table 4.1: Balance regimes represented by the full Matrix of Error.

Regime	Matrix or ratio condition	Scalar behavior
Strict zero equilibrium	$\mathcal{B} = \mathcal{C} = 0$, ρ undefined	The full Matrix of Error is zero and $E' = 0$.
Nonzero aggregate balance	$\mathcal{B} = \mathcal{C} > 0$, $\rho = 1$	Nonzero terms cancel instantaneously and $E' = 0$.
Dissipation-dominant	$\rho < 1$	$E' < 0$.
Stretching-dominant	$\rho > 1$	$E' > 0$.
Balance crossing	$\rho(t)$ crosses 1	The sign of E' changes while the matrix form remains fixed.

In the zero-mean periodic class used here, $\mathcal{C} = \nu \|\nabla\omega\|_{L^2}^2 = 0$ implies $\omega = 0$. Consequently, the strict-zero condition $\mathcal{B} = \mathcal{C} = 0$ makes both columns of the full Matrix of Error vanish, which is the equivalence used in the first row of Table 4.1.

Strict zero equilibrium and nonzero aggregate balance are therefore not the same physical situation. The first removes the operative terms. The second retains nonzero stretching and diffusion whose

scalar contributions happen to cancel at that instant. Dissipation dominance, stretching dominance, and crossings likewise describe outputs of one matrix rather than different governing models.

4.2 Exact Amplitude Construction and Fixed Design

The application reconstructs the Experiment 9 field generated from seed 20260911 on the periodic domain $[0, 2\pi)^3$, using a 24^3 grid, viscosity $\nu = 0.02$, and its state at time $t = 0.68$. The independently reconstructed reference ratio is

$$\rho_* = 0.955849373439646.$$

For a fixed divergence-free field u_*, set $u_\lambda = \lambda u_*$. Because vorticity and strain are linear in the field amplitude,

$$\mathcal{B}_\lambda = \lambda^3 \mathcal{B}_*, \qquad \mathcal{C}_\lambda = \lambda^2 \mathcal{C}_*, \qquad E_\lambda = \lambda^2 E_*, \qquad \rho_\lambda = \lambda \rho_*. \tag{4.4}$$

These exact laws place the four predetermined fields at the desired initial balance locations without fitting their subsequent trajectories.

Table 4.2: Fixed amplitude construction and observed initial ratio.

Case	λ	Observed $\rho(0)$	Initial regime
zero	0	undefined	strict zero equilibrium
below	0.7846424561	0.7500000000	dissipation-dominant
boundary	1.0461899415	1.0000000001	nonzero aggregate balance
above	1.3077374268	1.2500000000	stretching-dominant

Figure 4.1 shows the scaling continuously. At fixed spatial geometry, the cubic stretching contribution eventually overtakes the quadratic viscous contribution as amplitude increases. The zero case lies at the origin, while the second zero of $\mathcal{B}_\lambda - \mathcal{C}_\lambda$ is the nonzero aggregate-balance amplitude.

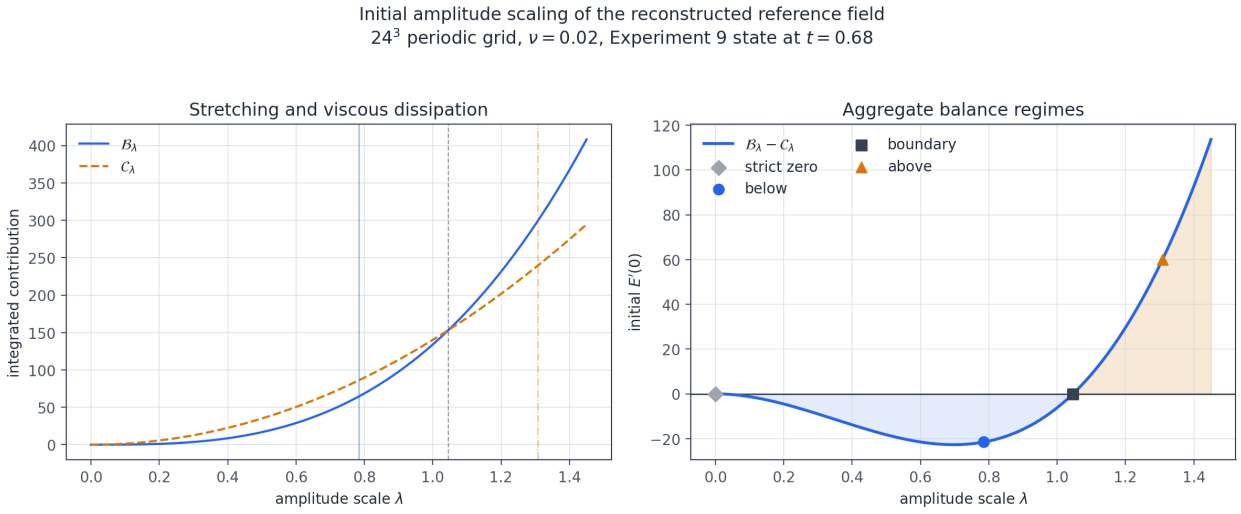


Figure 4.1: Exact initial amplitude scaling of the reconstructed field. The curves are construction identities; the marked cases are fixed inputs to the subsequent evolution.

The three nonzero fields were evolved sequentially to $t = 0.5$ with the same dealiased Fourier pseudospectral solver and fourth-order Runge–Kutta method, using $\Delta t = 0.005$ and 51 equally spaced saved times. The zero field was recorded analytically. One predetermined half-step run of the above case used $\Delta t = 0.0025$. No search, optimization, coefficient fitting, noise injection, library change, or adaptive extension was performed.

All integrity conditions passed. The maximum full-scalar closure error was 5.06×10^{-16}, the independent enstrophy-rate error was 4.30×10^{-16}, and divergence remained below 6.00×10^{-16}. The maximum CFL number was 0.0409. At the 51 matched times, the half-step run differed from the primary above run by at most 2.37×10^{-10} in normalized enstrophy and 2.46×10^{-10} relatively in ρ. An independent audit of the saved tables also reconstructed every aggregate and block identity.

4.3 Observed Regime Paths

Table 4.3 reports the results without assigning a preferred trajectory shape. The three nonzero ratios decline over the sampled interval, but they begin in different regimes and consequently produce different enstrophy histories.

Table 4.3: Observed scalar-balance and normalized-enstrophy paths.

Case	$\rho(0)$	$\rho(0.5)$	max. E/E_0	final E/E_0	Crossing result
below	0.750000	0.679818	1.000000	0.934690	none
boundary	1.000000	0.845740	1.000000	0.982692	leaves balance immediately
above	1.250000	0.974170	1.034606	1.034245	$\rho = 1$ at $t = 0.458898$

The below case remains dissipation-dominant and loses approximately 6.53% of its initial enstrophy. The boundary case begins with nonzero stretching and dissipation in instantaneous balance, but that cancellation is not a stationary field: it moves into dissipation dominance and loses approximately 1.73%. The above case exhibits finite transient stretching-driven growth, reaches a 3.46% increase, and then turns downward after crossing the balance line. It still ends 3.42% above its initial enstrophy at the fixed stopping time.

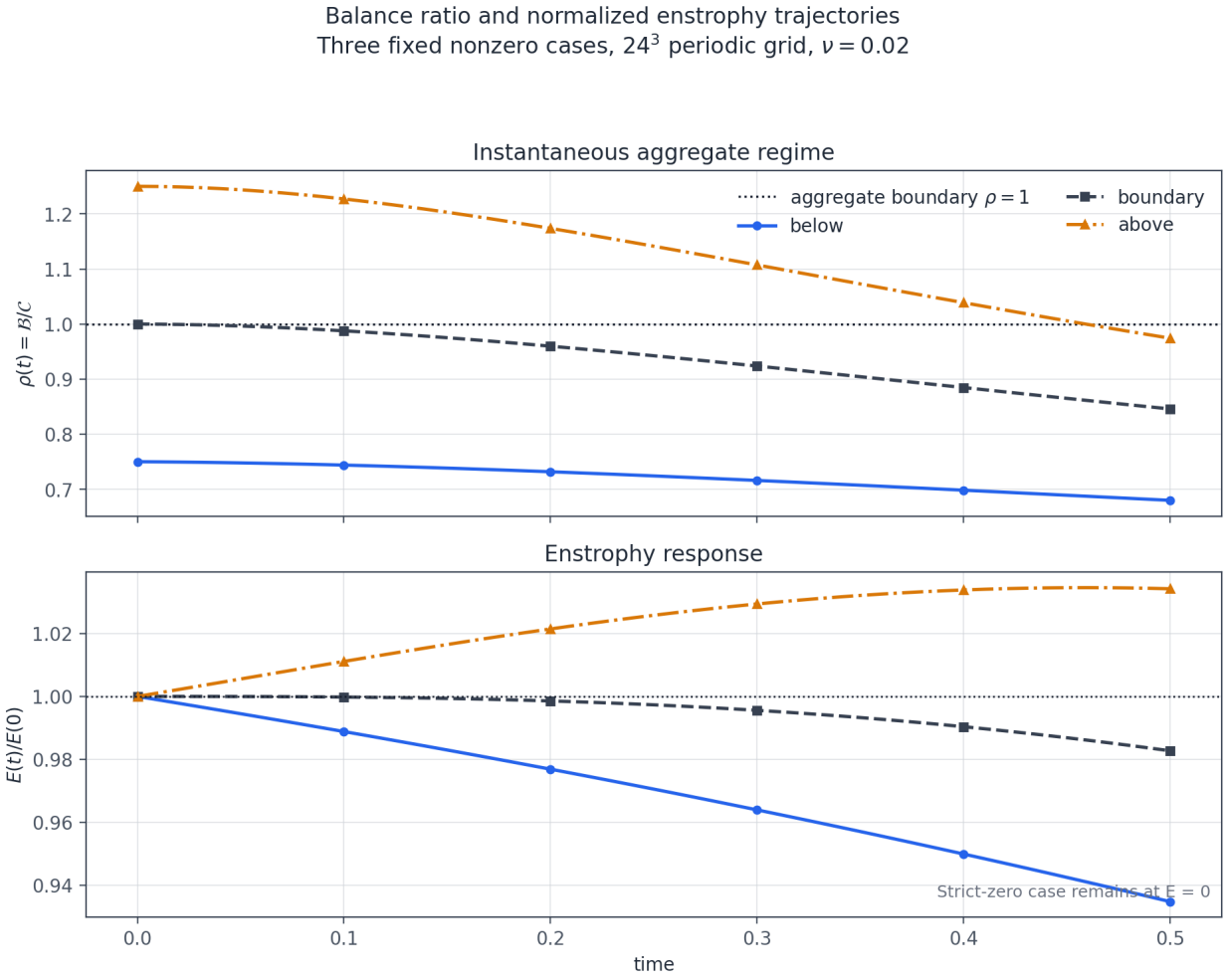


Figure 4.2: Observed balance ratios and normalized-enstrophy responses for the three fixed nonzero cases. No trajectory direction was specified before the calculation.

Figure 4.3 removes time from one axis and shows the same result in the $(\mathcal{C}, \mathcal{B})$ plane. The below path stays beneath the diagonal. The boundary path departs the diagonal toward dissipation dominance. The above path begins above it and crosses near the end of the fixed interval. The origin records strict zero equilibrium and remains separate from nonzero aggregate balance.

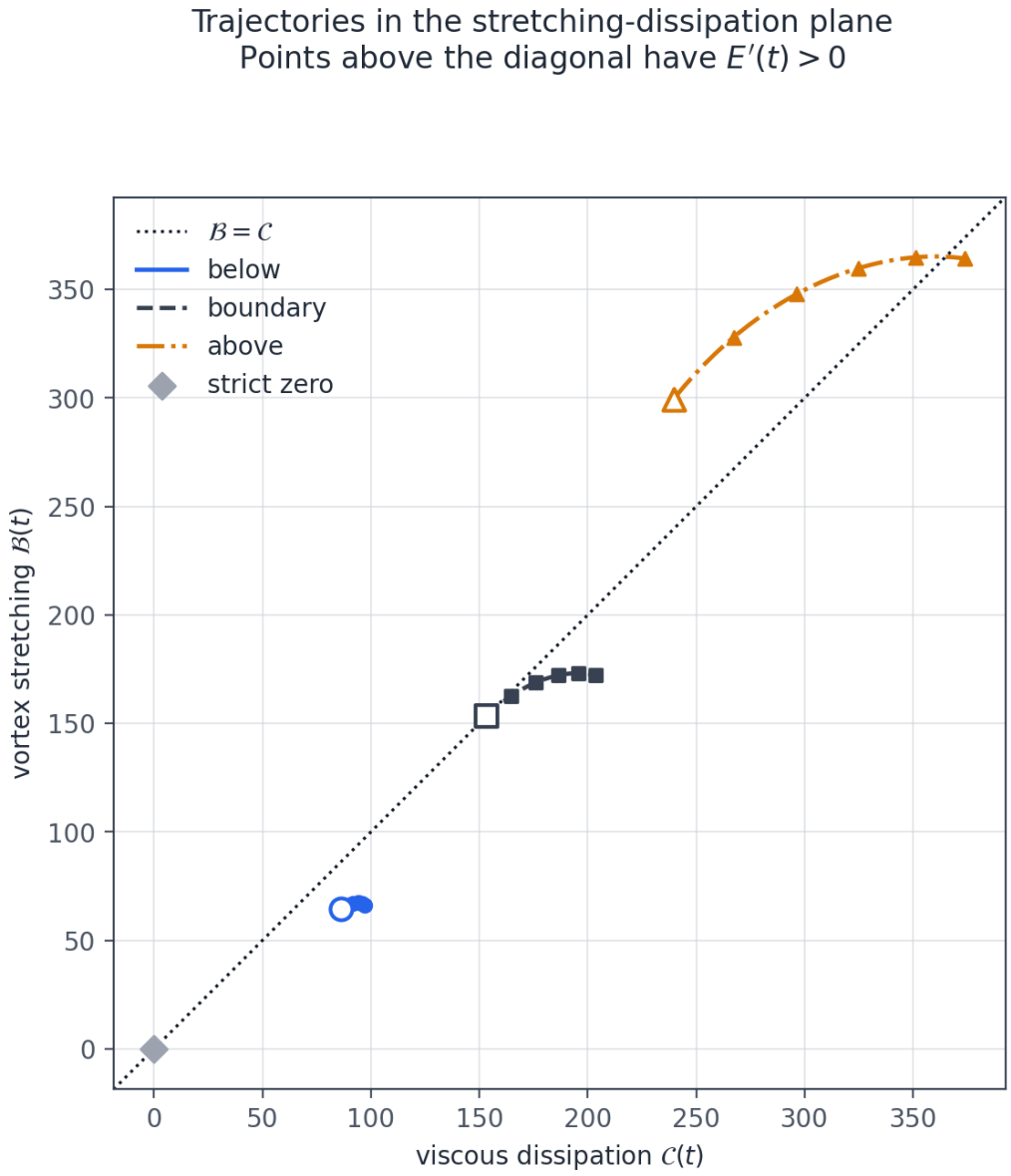


Figure 4.3: Trajectories in the stretching–dissipation plane. Open markers show the initial nonzero states; the diagonal is instantaneous aggregate balance.

4.4 Matrix-Block Behavior

The scalar paths retain the full coefficient structure derived in Part II. With one coordinate isolated,

$$\omega = \begin{bmatrix} \eta \\ \zeta \end{bmatrix}, \qquad S = \begin{bmatrix} B & b \\ b^{\mathsf{T}} & s \end{bmatrix},$$

so the integrated stretching contribution is

$$\mathcal{B} = \underbrace{\int \eta^{\mathsf{T}} B \eta \, \mathrm{d}x}_{\mathcal{B}_{2\times 2}} + \underbrace{\int 2\zeta b^{\mathsf{T}} \eta \, \mathrm{d}x}_{\mathcal{B}_{\mathrm{cross}}} + \underbrace{\int s\zeta^2 \, \mathrm{d}x}_{\mathcal{B}_{\mathrm{scalar}}}. \tag{4.5}$$

Writing $\mathcal{C}_2$ and $\mathcal{C}_1$ for the positive magnitudes of the retained-coordinate and isolated-coordinate viscous contributions gives

$$E' = \mathcal{B}_{2\times 2} + \mathcal{B}_{\mathrm{cross}} + \mathcal{B}_{\mathrm{scalar}} - \mathcal{C}_2 - \mathcal{C}_1. \tag{4.6}$$

Figure 4.4 displays all five terms. In every case the cross block $\mathcal{B}_{\mathrm{cross}}$ is the largest positive stretching component at the time of maximum aggregate rate. For the below, boundary, and above cases its corresponding values are 29.0871, 68.9472, and 143.8547. The above case reaches its largest aggregate rate at $t = 0.06$, where total stretching is 317.6136, total dissipation is 255.9085, and the resulting rate is 61.7051. The full five-term sum agrees with the direct scalar calculation to within 5.06×10^{-16} relatively.

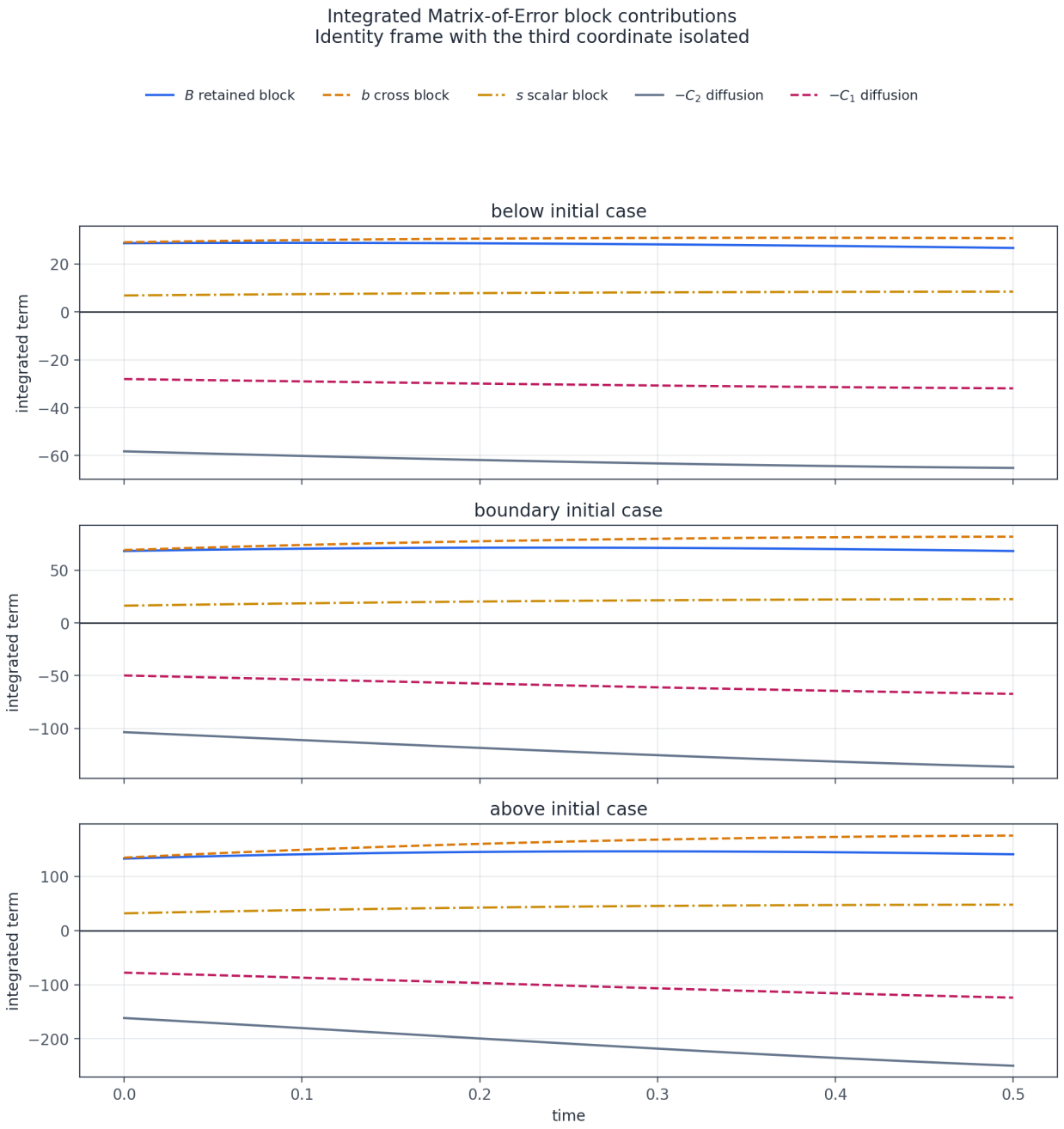


Figure 4.4: Integrated contributions of the three stretching blocks and two viscous blocks in one fixed coordinate split. The signs displayed are their signs in the aggregate enstrophy rate.

The block histories make the physical distinction visible. Aggregate decay does not mean that stretching has disappeared, and aggregate growth does not mean that viscosity has disappeared. Both mechanisms remain active. Their five contributions determine whether the instantaneous scalar output lies below, on, or above the balance boundary.

4.5 Smooth-Continuation Display

Section 3.7 proposed the smooth-continuation form

$$\mathcal{B}(t) \le (1-\delta)\mathcal{C}(t) + g(t)\,\|\omega(t)\|_{L^2}^2\,, \qquad \delta > 0. \tag{4.7}$$

The continuation quantities have distinct roles. The notation $\overline{g}_{u_0,\nu}$ denotes a majorant with the cumulative control specified in Section 5; $g_{\mathrm{std}}[\omega]$ denotes the standard full-vorticity majorant; and $g_{\mathrm{blk},\chi}$ is the selected blockwise majorant. The least nonnegative remainder calculated from a completed trajectory is g_{req}. Only g_{req} is evaluated in this subsection. Their definitions and the full estimate are collected in Sections 5.4–5.8.

For this application, keep the Experiment 9 margin $\delta = 0.0441506266$ fixed and evaluate the least nonnegative pointwise remainder in equation (5.10) along each displayed path. This is a diagnostic extracted from the completed matrix, not a new physical term. This subsection is a specific demonstration of the derived form, so the computed remainder and envelope inherently retain the same fundamental shape as the equation that defines the demonstration. Substitution into the enstrophy identity gives the displayed envelope

$$E(t) \le E(0)\exp\!\left(2\int_0^t g_{\mathrm{req}}(s)\;\mathrm{d}s\right). \tag{4.8}$$

The below path requires no remainder over the sampled interval: $g_{\mathrm{req}} = 0$. The boundary path has maximum $g_{\mathrm{req}} = 0.009627$ and accumulated integral 0.001349. The above path has maximum 0.065962 and accumulated integral 0.022900. Their final normalized envelopes are respectively 1.000000, 1.002703, and 1.046865; every measured path remains below its corresponding envelope. The finite stretching-dominant interval is therefore represented inside the same continuation display rather than treated as an exception to the matrix.

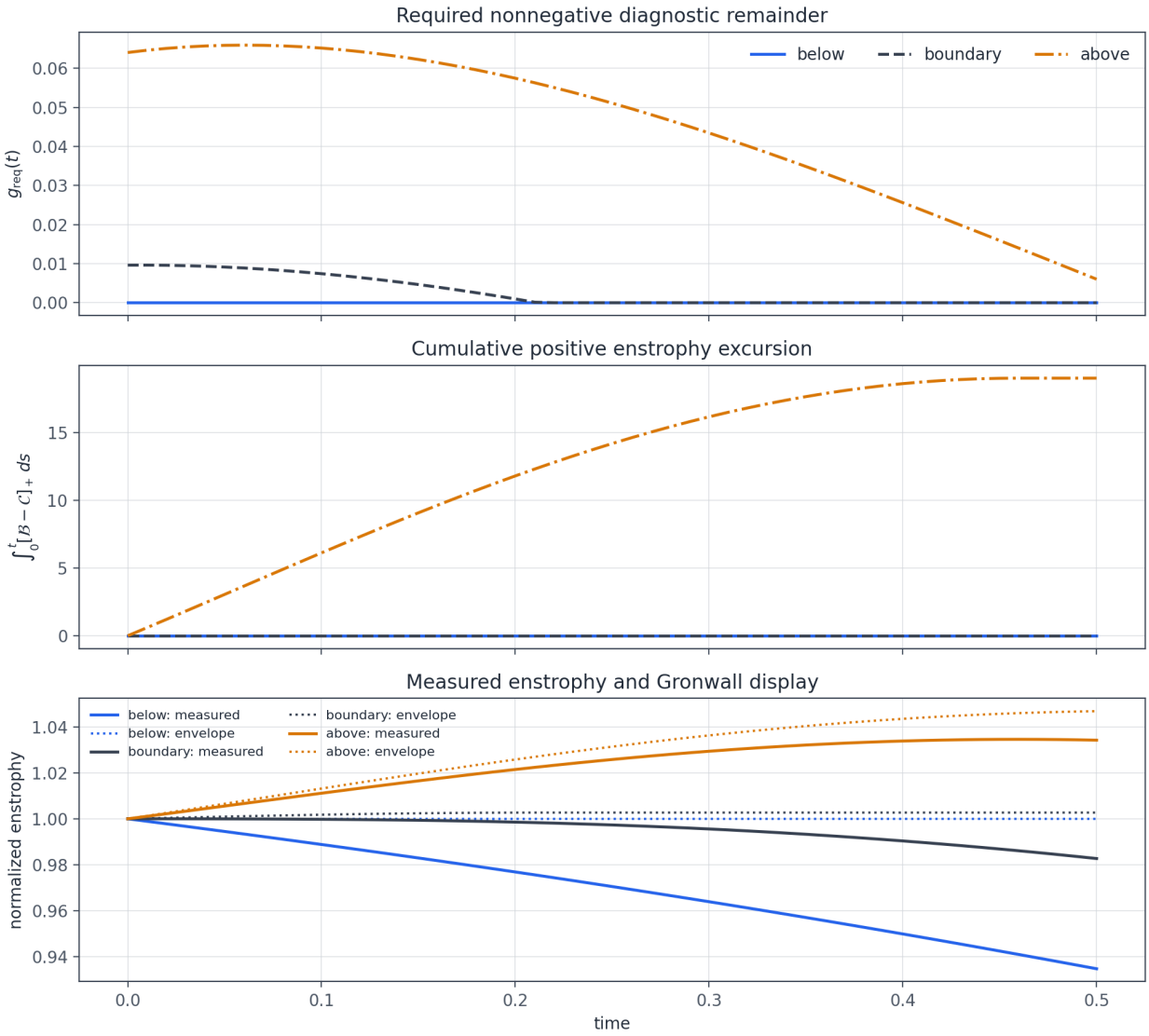


Figure 4.5: The fixed-margin diagnostic remainder, cumulative positive enstrophy excursion, and measured enstrophy beside the corresponding Gronwall display.

4.6 Physical Interpretation

The amplitude atlas supplies the first physical reading. For one field geometry, increasing amplitude strengthens integrated vortex stretching as λ^3 while viscous dissipation grows as λ^2. A sufficiently small realization is therefore diffusion-led at the initial instant, while a larger realization can be stretching-led. This conclusion comes directly from the homogeneity of the two established Navier–Stokes terms.

The trajectories supply the second reading: initial balance does not fix later balance. The evolving

velocity gradients, vorticity, and spectral content change both contributions. In this recorded field, all three nonzero ratios move downward. The below and boundary cases consequently decay, while the above case undergoes transient vorticity amplification before viscosity overtakes stretching. The crossing is ordinary represented dynamics, not a change of equation or a failure of the Matrix of Error.

The blocks supply the third reading. Cross-coordinate stretching is not a small decorative correction in this realization; it is the largest positive stretching block at each case's maximum aggregate rate. At the same time, both viscous blocks remain active. The observed scalar behavior is produced by their full competition, which is why the complete matrix remains the appropriate applications object.

Finally, the origin and the nonzero boundary show two different meanings of equilibrium. The origin is the strict non-error state proposed by omni-dimensional equilibrium. The boundary field has active nonzero terms and only an instantaneous scalar cancellation. Its immediate departure from the diagonal demonstrates why a zero aggregate cannot be substituted for a zero matrix.

4.7 Result and Significance

Experiment 10 provides the planned graphical application of the completed three-dimensional Navier–Stokes Matrix of Error. One unchanged representation displays strict zero equilibrium, nonzero aggregate balance, dissipation-dominant decay, stretching-dominant growth, and a transition between the nonzero regimes. The full block structure remains numerically exact in every application, and the fixed smooth-continuation form accommodates both decreasing enstrophy and finite transient growth.

Complex-valued spectral representation does not create a separate physical route around this balance. Its imaginary components encode phase and oscillation, while physical amplitude growth remains governed by the real stretching–dissipation balance represented by the matrix. Sustained input from external forcing or additional physical mechanisms would enter as further system-specific contributions. Those contributions lie outside the unforced periodic application considered here, but remain expressible through the shared nonlinear-calculus map when they belong to the supplied physical system.

The result is deliberately descriptive: no trajectory direction was proposed before execution, and no result was edited to match an external expectation. Its value is to make the completed mathematical result physically legible. The previous sections supply the cross-system support, derivation, test, and proposed smooth-continuation interpretation. Section 4 shows the range of behaviors produced when that result is used. Section 5 connects these applications to the complete visible chain of identities, majorants, and smooth-continuation implications.

5 Smooth-Continuation Interpretation: The Full Visible Chain

This section supplies the full visible chain for the paper's Smooth-Continuation Interpretation. Section 2 provides the derived Matrix of Error and its exact scalar identity; Section 3 provides the completed full-matrix validation and observed dissipative margin; and Section 4 displays zero and nonzero continuation remainders in the completed applications. The purpose here is to put their mathematical relationship, the selected majorant, and the resulting claim in one continuous explanation.

5.1 Specific Claim and Mathematical Setting

The specific analytic claim is that the full three-dimensional Matrix of Error yields an explicit blockwise majorant of the positive stretching surplus. Its time integral controls the enstrophy through the displayed Gronwall estimate, and finite cumulative control through a finite horizon implies smooth continuation through that horizon. The universal interpretation advanced by nonlinear calculus is that this cumulative control holds for every admissible initial field and every finite horizon. The majorant estimate and the continuation implication are proved below; the universal integrability assertion is stated explicitly as the theory's claim.

All analytic statements in this section concern the unforced incompressible Navier–Stokes system on the three-dimensional periodic domain Ω, with viscosity $\nu > 0$ and smooth divergence-free initial velocity u_0. Let $[0, T_*)$ be its maximal smooth interval. The notation T_* does not assert a breakdown: $T_* = \infty$ denotes global smooth existence. A finite $T > 0$ denotes the horizon considered in the estimate. Spatial norms and integrals are over Ω, unless another domain is specified.

Write

$$\omega = \nabla \times u, \qquad S = \frac{1}{2}(\nabla u + \nabla u^{\mathsf{T}}), \qquad Y(t) = \|\omega(t)\|_{L^2}^2 = 2E(t). \tag{5.1}$$

Thus Y is twice the enstrophy E plotted in Section 4; normalized $Y/Y(0)$ and $E/E(0)$ are identical. The matrix block B and the scalar integral $\mathcal{B}$ retain their distinct meanings. The constants C_Ω and $C_{\Omega,\delta}$ are analytic estimate constants, not fitted physical coefficients. They may be enlarged when combining estimates.

Table 5.1: How the completed sections enter the Smooth-Continuation Interpretation.

Source	Established content used here	Role in the visible chain
Section 2	Full matrix, exact two-plus-one form, and enstrophy identity	Determines the complete signed stretching and diffusion balance.
Section 3	Full reconstruction and observed dissipative margin	Supplies the tested zero-remainder realization.
Section 4	Fixed applications, positive remainders, and plotted envelopes	Displays decay, transient growth, and crossings within the same form.
Section 5	Explicit majorants and conditional continuation theorem	States the analytic implication and the universal interpretation together.

This organization keeps the experimental completion and the mathematical interpretation explicit. The data in the preceding sections are the entire empirical basis used here.

5.2 From the Full Matrix to the Exact Enstrophy Identity

The starting point is the established periodic, unforced system

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = u_0. \tag{5.2}$$

Taking the curl eliminates the pressure gradient. Incompressibility gives the vorticity equation

$$\frac{D\omega}{Dt} = (\omega \cdot \nabla)u + \nu \Delta \omega = S\omega + \nu \Delta \omega = \mathfrak{E}_{\mathrm{NS}} \mathbf{1}_2, \qquad \mathfrak{E}_{\mathrm{NS}} = [S\omega \mid \nu \Delta \omega]. \tag{5.3}$$

Here $D/Dt = \partial_t + u \cdot \nabla$. The antisymmetric part of ∇u annihilates ω, giving $(\nabla u)\omega = S\omega$. This is the pointwise full matrix derived in Section 2.4, with the transport contribution retained in the material derivative.

Take the L^2 inner product of (5.3) with ω. The transport contribution is

$$\int_\Omega \omega \cdot (u \cdot \nabla)\omega \, \mathrm{d}x = \frac{1}{2} \int_\Omega u \cdot \nabla |\omega|^2 \, \mathrm{d}x = 0, \tag{5.4}$$

by periodicity and $\nabla \cdot u = 0$. Periodic integration by parts gives

$$\nu \int_\Omega \omega \cdot \Delta \omega \, \mathrm{d}x = -\nu \, \|\nabla \omega\|_{L^2}^2 \, . \tag{5.5}$$

Consequently,

$$\boxed{\frac{1}{2} Y'(t) = \mathcal{B}(t) - \mathcal{C}(t), \qquad \mathcal{B}(t) = \int_\Omega \omega^{\mathsf{T}} S \omega \, \mathrm{d}x, \qquad \mathcal{C}(t) = \nu \, \|\nabla \omega\|_{L^2}^2 \, .} \tag{5.6}$$

These are exact identities for smooth solutions [16, 17, 18]. Experiment 9 checks their numerical realization independently, including agreement between the solver's enstrophy rate and the reconstructed stretching–diffusion balance.

The full two-plus-one decomposition retains

$$\omega = \begin{bmatrix} \eta \\ \zeta \end{bmatrix}, \qquad S = \begin{bmatrix} B & b \\ b^{\mathsf{T}} & s \end{bmatrix}, \qquad \nu \Delta \omega = \begin{bmatrix} C_2 \\ C_1 \end{bmatrix}.$$

It therefore gives

$$\begin{aligned} \omega^{\mathsf{T}} \mathfrak{E}_{\mathrm{NS}} \mathbf{1}_2 = \eta^{\mathsf{T}} B \eta + 2\zeta b^{\mathsf{T}} \eta + s\zeta^2 \\ + \eta^{\mathsf{T}} C_2 + \zeta C_1. \end{aligned} \tag{5.7}$$

The cross term $2\zeta b^{\mathsf{T}} \eta$ and the isolated-direction term $s\zeta^2$ are thus present before any inequality is applied. Their omission is the naive-form diagnostic tested in Section 3. Every estimate below acts on the full expression (5.7).

5.3 The Observed Margin and the Positive Remainder

Fix $0 < \delta < 1$. Retaining the fraction δ of the viscous contribution gives the surplus

$$R_\delta(t) = \mathcal{B}(t) - (1 - \delta)\mathcal{C}(t). \tag{5.8}$$

Its exact matrix-block expression is

$$R_\delta(t) = \int_\Omega \big(\eta^{\mathsf{T}} B\eta + 2\zeta b^{\mathsf{T}}\eta + s\zeta^2 + (1-\delta)(\eta^{\mathsf{T}} C_2 + \zeta C_1)\big)\,\mathrm{d}x. \tag{5.9}$$

The negative integrated diffusion terms account for the plus sign on the second line. Equation (5.9) is the full matrix balance with a retained dissipative margin.

For $Y(t) > 0$, define the least nonnegative coefficient needed to bound this surplus:

$$g_{\mathrm{req}}(t) = \frac{[R_\delta(t)]_+}{Y(t)}, \qquad [r]_+ = \max\{r, 0\}. \tag{5.10}$$

Set $g_{\mathrm{req}} = 0$ in the zero-vorticity state. That convention is consistent with the matrix: zero vorticity gives zero stretching and diffusion. For any nonnegative g, the following equivalence is exact:

$$\mathcal{B} \le (1-\delta)\mathcal{C} + gY \quad \Longleftrightarrow \quad g \ge g_{\mathrm{req}} \qquad (Y > 0). \tag{5.11}$$

Thus the definition specifies precisely which quantity the proposed continuation form controls.

There are two different scalar thresholds. The physical growth threshold is $\rho = \mathcal{B}/\mathcal{C} = 1$, when $\mathcal{C} > 0$. The zero-remainder threshold is $\rho = 1 - \delta$. More explicitly,

$$E' = \mathcal{C}(\rho - 1), \qquad g_{\mathrm{req}} = \frac{\mathcal{C}}{Y}[\rho - (1-\delta)]_+. \tag{5.12}$$

A positive g_{req} can therefore occur during decay as well as growth: the interval $1 - \delta < \rho \le 1$ has nonpositive enstrophy rate but requires a positive remainder to retain the chosen margin. This explains the positive remainder in the nonzero boundary application of Section 4.

Experiment 9 supplies the stronger zero-remainder case. At the discrete sample level, defining

$$\delta^{24}_{\mathrm{emp}} = 1 - \max_{\text{primary samples}} \mathcal{B}/\mathcal{C} \simeq 0.0441506266$$

makes $g_{\mathrm{req}} = 0$ throughout that primary record; the printed decimal values have finite precision. Equation (3.8) records the resulting negative enstrophy rate. The 32^3 comparison has its own observed margin, reported in Section 3. These observations corroborate the exact surplus construction without changing its definition.

5.4 The Standard Estimate and the Roles of the Majorants

For a periodic vorticity field, its spatial mean is zero because it is a curl. The periodic Biot–Savart estimate controls S by ω in L^3, and the mean-zero interpolation inequality gives

$$\|\omega\|_{L^3} \le C_\Omega \|\omega\|_{L^2}^{1/2} \|\nabla\omega\|_{L^2}^{1/2}.$$

Hölder's inequality, interpolation, and Young's inequality therefore yield

$$\begin{aligned} \mathcal{B} \le |\mathcal{B}| &\le \|S\|_{L^3} \|\omega\|_{L^3}^2 \le C_\Omega \|\omega\|_{L^3}^3 \\ &\le C_\Omega \|\omega\|_{L^2}^{3/2} \|\nabla\omega\|_{L^2}^{3/2} \\ &\le (1-\delta)\nu \|\nabla\omega\|_{L^2}^2 + C_{\Omega,\delta}\nu^{-3} \|\omega\|_{L^2}^6. \end{aligned} \tag{5.13}$$

Young's conjugate exponents are 4/3 and 4; absorbing the gradient factor into $(1-\delta)\nu\,\|\nabla\omega\|_{L^2}^2$ produces the factor ν^{-3} in the remaining term. This is the standard solution-dependent estimate [16, 17, 18].

It supplies the explicit majorant

$$g_{\mathrm{std}}[\omega](t) = C_{\Omega,\delta}\nu^{-3}\,\|\omega(t)\|_{L^2}^4\,, \qquad 0 \le g_{\mathrm{req}} \le g_{\mathrm{std}}[\omega]. \tag{5.14}$$

The inequality form is therefore already derived on the smooth interval. The coefficient is computed from the evolving solution, which is compatible with its role as a pointwise estimate.

Table 5.2: The four control quantities and their distinct roles.

Quantity	Meaning in the paper
g_{req}	Least nonnegative surplus coefficient; evaluated on the completed trajectories in Section 4.
$g_{\mathrm{std}}[\omega]$	Analytic majorant obtained from the standard full-vorticity estimate.
$g_{\mathrm{blk},\chi}$	Selected analytic majorant obtained from the retained two-component block.
$\overline{g}_{u_0,\nu}$	Notation for a majorant together with the cumulative control specified in the continuation statement.

A computed value of g_{req} is not a measurement of either analytic estimate constant. In particular, $g_{\mathrm{req}} = 0$ does not force $g_{\mathrm{blk},\chi} = 0$: a valid upper majorant may be positive even when the signed surplus is nonpositive. This distinction matters when connecting the experiments to the analytic form.

5.5 The Selected Blockwise Majorant

Fix an arbitrary unit direction $\chi \in \mathbb{R}^3$, with $|\chi| = 1$, constant in space and time, and define

$$P_\chi = I - \chi\otimes\chi, \qquad \eta_\chi = P_\chi\omega, \qquad \omega'_\chi = (\chi\cdot\omega)\chi, \qquad \omega = \eta_\chi + \omega'_\chi. \tag{5.15}$$

In the coordinate split with χ isolated, η_χ is the retained two-component vorticity block. Both components and the isolated direction remain in the full matrix. The estimate uses the two-component norm to control the aggregate stretching; it does not remove the other stretching blocks.

The fixed-plane singular-integral decomposition of the stretching contribution gives

$$|\mathcal{B}| \le C_\Omega\,\|\eta_\chi\|_{L^2}\,\|\omega\|_{L^4}^2\,. \tag{5.16}$$

This is the periodic counterpart of the two-component estimate of Chae and Choe [22]. In their decomposition, the singular-integral terms retain at least one factor from the two-component vorticity. The Calderón–Zygmund bounds and Hölder's inequality put that factor in L^2 and the remaining two factors in L^4. On the periodic domain, the corresponding mean-zero singular-integral bounds supply the constants in (5.16). The estimate acts after spatial integration; it does not assert a pointwise sign or cancellation of each matrix block.

The periodic interpolation estimate

$$\|\omega\|_{L^4}^2 \le C_\Omega\,\|\omega\|_{L^2}^{1/2}\,\|\nabla\omega\|_{L^2}^{3/2}$$

and Young's inequality now give the complete blockwise chain:

$$\begin{aligned}\mathcal{B} \le |\mathcal{B}| &\le C_\Omega \|\eta_\chi\|_{L^2} \|\omega\|_{L^4}^2 \\ &\le C_\Omega \|\eta_\chi\|_{L^2} \|\omega\|_{L^2}^{1/2} \|\nabla\omega\|_{L^2}^{3/2} \\ &\le (1-\delta)\mathcal{C} + C_{\Omega,\delta}\nu^{-3} \|\eta_\chi\|_{L^2}^4 Y.\end{aligned} \tag{5.17}$$

The selected majorant is consequently

$$\boxed{g_{\mathrm{blk},\chi}(t) = C_{\Omega,\delta}\nu^{-3} \|P_\chi\omega(t)\|_{L^2}^4 .} \tag{5.18}$$

Equation (5.17) proves that this function majorizes the required surplus. Enlarging the constant so the same $C_{\Omega,\delta}$ works in the standard and blockwise estimates yields

$$0 \le g_{\mathrm{req}} \le g_{\mathrm{blk},\chi} \le g_{\mathrm{std}}[\omega], \tag{5.19}$$

where the last inequality uses $\|P_\chi\omega\|_{L^2} \le \|\omega\|_{L^2}$.

This fixes the analytic choice in the interpretation. It uses the two-plus-one geometry already present in the derived Matrix of Error and replaces the full-vorticity norm in the standard estimate by the two-component norm. The selected function and its pointwise majorization are established by the displayed calculation.

5.6 Comparison with Chae–Choe and Miller

Chae and Choe [22, Theorem 1] prove regularity on $\mathbb{R}^3$ under control of two fixed vorticity components. Their criterion includes

$$\eta_\chi \in L_t^p L_x^q, \qquad \frac{2}{p} + \frac{3}{q} = 2, \qquad \frac{3}{2} < q < \infty.$$

The choice $p = 4,\ q = 2$ is exactly the norm used in the selected blockwise majorant. Their proof controls stretching by singular-integral estimates, absorbs a gradient term into viscosity, and applies Gronwall's inequality. Equation (5.17) uses that same analytic structure in the periodic setting.

Miller [23] expresses the fixed-plane criterion through $\chi \times \omega$, whose magnitude equals $|P_\chi\omega|$. His variable-direction theorem allows a unit vector $v(x,t)$, assumes $\nabla v \in L^\infty_{t,\mathrm{loc}} L^\infty_x$, and uses the strain formulation to obtain regularity from $v \times \omega \in L_t^4 L_x^2$. The extension permits the controlled plane to vary in space and time. The present choice is fixed-direction, so its criterion corresponds to the Chae–Choe case rather than to Miller's additional variable-direction generality. Both cited theorems are stated on $\mathbb{R}^3$; this paper's estimates use the periodic operators on Ω.

The exponent 4 has a specific mathematical origin. Under the Navier–Stokes scaling on $\mathbb{R}^3$,

$$u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t), \qquad \omega_\lambda(x,t) = \lambda^2 \omega(\lambda x, \lambda^2 t),$$

the mixed vorticity norm scales with exponent $2 - 3/q - 2/p$. Thus $2/p + 3/q = 2$ makes it scale invariant. At $q = 2$, this gives $p = 4$. The fourth power in (5.18) also follows directly from Young's inequality in the displayed blockwise derivation.

These are conditional regularity theorems, not impossibility results. They establish the consequence of controlling a specified quantity and do not prohibit establishing that control. The criterion itself is prior mathematics. The connection made here is its realization in the full Matrix-of-Error framework and its explicit relation to the completed cross-system derivation, three-dimensional validation, and applications.

5.7 Cancellation and Geometric Depletion within the Selected Interpretation

The earlier route comparison concerns three ways of retaining or estimating information in the same full stretching contribution. The blockwise majorant in (5.18) is the selected control quantity. Cancellation and geometric depletion explain related structure; no additional route is selected here.

5.7.1 Exact cancellation

For periodic incompressible fields, the Betchov identity is

$$4\int_\Omega \operatorname{tr}(S^3)\,\mathrm{d}x = -3\int_\Omega \omega^{\mathsf{T}} S\omega\,\mathrm{d}x = -3\mathcal{B}(t). \tag{5.20}$$

It is an exact integrated relation [19]. In the retained two-plus-one blocks,

$$\operatorname{tr}(S^3) = \operatorname{tr}(B^3) + 3b^{\mathsf{T}}Bb + 3s|b|^2 + s^3, \qquad \operatorname{tr}(B) + s = 0. \tag{5.21}$$

These formulas preserve cancellation among signed contributions before an absolute-value estimate is taken. They do not assign a fixed sign to $\mathcal{B}$, and they do not change the selected majorant. Their mathematical role is an exact representation of the same integrated stretching.

5.7.2 Geometric depletion

Where $\omega \neq 0$, write $\xi = \omega/|\omega|$. The singular-integral representation of vortex stretching contains the relative orientation of vorticity at different points. Established direction-coherence results show how suitably controlled variation in ξ in high-vorticity regions suppresses effective stretching and yields regularity [20, 21]. In the present notation, an estimate of this kind would be expressed as

$$[R_\delta(t)]_+ \le g_{\mathrm{geo}}(t)Y(t) \tag{5.22}$$

together with finite cumulative control of g_{geo}. This describes the mathematical role of the geometric criterion in the comparison.

Choosing $\chi = \xi(x,t)$ pointwise makes $P_\chi\omega = 0$ algebraically, but it changes a fixed projection into a spatially varying one. The fixed-direction singular-integral estimate used in (5.16) does not make that substitution. Miller's hypothesis on ∇v is precisely the kind of additional geometric control that accompanies a moving direction. The cancellation, blockwise, and geometric statements therefore carry different information. The interpretation in this paper uses the already selected fixed-direction blockwise majorant.

5.8 The Smooth-Continuation Estimate and Its Exact Implication

Let $g \ge 0$ be locally integrable on $[0, T_*)$ and satisfy $g \ge g_{\mathrm{req}}$. The matrix balance and (5.11) immediately give

$$\mathcal{B} \le (1-\delta)\mathcal{C} + gY, \qquad Y' + 2\delta\mathcal{C} \le 2gY. \tag{5.23}$$

Define the accumulated coefficient and integrating factor by

$$A_g(t) = \int_0^t g(s)\,\mathrm{d}s, \qquad H_g(t) = \exp(-2A_g(t)). \tag{5.24}$$

Since $H_g' = -2gH_g$ almost everywhere, multiplication of (5.23) by H_g yields

$$(H_g Y)' + 2\delta H_g \mathcal{C} \le 0.$$

Integrating from 0 to $t < T_*$ gives the complete weighted estimate

$$H_g(t)Y(t) + 2\delta \int_0^t H_g(s)\mathcal{C}(s)\ \mathrm{d}s \le Y(0). \tag{5.25}$$

In particular,

$$\boxed{Y(t) \le Y(0)\exp(2A_g(t)).} \tag{5.26}$$

The same estimate holds for $E = Y/2$, so it is the mathematical form used in the Section 4 display.

Substituting the selected majorant fixes the visible chain:

$$\boxed{Y(t) \le Y(0)\exp\left(2C_{\Omega,\delta}\nu^{-3}\int_0^t \|P_\chi \omega(s)\|_{L^2}^4\ \mathrm{d}s\right).} \tag{5.27}$$

No choice among the earlier routes remains in this expression.

Proposition: smooth continuation under finite cumulative control. Let u_0 be smooth, periodic, and divergence-free, let $\nu > 0$, and let u be the associated maximal smooth solution on $[0, T_*)$. Fix $0 < \delta < 1$. Suppose $g \ge g_{\mathrm{req}}$, $g \ge 0$, and for a prescribed finite $T > 0$,

$$\sup_{0<t<\min\{T,T_*\}} A_g(t) \le A_T < \infty. \tag{5.28}$$

Then $T_* > T$. In particular, this conclusion holds for $g = g_{\mathrm{blk},\chi}$ whenever its blockwise integral satisfies (5.28). If the condition holds for every finite T, the solution is global and smooth.

Proof. Equation (5.25) and (5.28) give

$$Y(t) \le Y(0)e^{2A_T}, \qquad \int_0^t \mathcal{C}(s)\ \mathrm{d}s \le \frac{Y(0)e^{2A_T}}{2\delta}, \quad t < \min\{T, T_*\}. \tag{5.29}$$

The second bound follows because $H_g(s) \ge e^{-2A_T}$ and both terms on the left side of (5.25) are nonnegative. The unforced velocity energy identity gives

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu\int_0^t \|\nabla u(s)\|_{L^2}^2\ \mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2. \tag{5.30}$$

For periodic divergence-free fields, $\|\nabla u\|_{L^2}^2 = \|\omega\|_{L^2}^2 = Y$. Thus (5.29) and (5.30) give a uniform H^1 bound on $u(t)$ before $\min\{T, T_*\}$.

The standard local strong-solution theory supplies a positive existence interval controlled by this H^1 bound, ν, and the periodic domain [16, 17]. If $T_* \le T$ were finite, restarting sufficiently close to T_* would therefore extend the strong solution beyond T_*. Uniqueness identifies the restarted solution with the existing one on their overlap, contradicting maximality. Smooth initial data and parabolic regularity give smooth continuation. Hence $T_* > T$. Applying the same conclusion for each finite T gives $T_* = \infty$. □

The proposition gives a complete implication with a specific hypothesis and conclusion. For the selected blockwise majorant, finite cumulative control is sufficient for the stated global smoothness conclusion when it holds on every finite horizon. Pointwise dissipation dominance is not a hypothesis of the proposition.

5.9 Initial-Data Bounds, Solution-Dependent Bounds, and the Universal Claim

The function $g_{\mathrm{blk},\chi}$ and its pointwise majorization are fixed by (5.18) and (5.19). Its accumulated integral is the quantity appearing in the proved exponential estimate. The universal integrability statement advanced in the interpretation is

$$\sup_{0<\tau<\min\{T,T_*\}} \int_0^\tau \|P_\chi \omega(t)\|_{L^2}^4 \, \mathrm{d}t \le F_\Omega(T,\nu,\|u_0\|_{H^1}) < \infty \tag{5.31}$$

for every admissible smooth initial field and every finite T. The fixed direction χ is selected independently of the future solution. A bound of the displayed form supplies $A_T = C_{\Omega,\delta}\nu^{-3}F_\Omega$ in the proposition.

More generally, the notation used in (3.9) can be written as

$$\begin{aligned} 0 \le g_{\mathrm{req}}(t) &\le \overline{g}_{u_0,\nu}(t), \\ \sup_{0<\tau<\min\{T,T_*\}} \int_0^\tau \overline{g}_{u_0,\nu}(t) \, \mathrm{d}t &\le G_\Omega(T,\nu,u_0) < \infty. \end{aligned} \tag{5.32}$$

The labels F_Ω and G_Ω specify upper bounds on accumulated quantities. They do not by definition supply numerical values or a proof of their finiteness.

5.9.1 What a priori means in this statement

An a priori estimate bounds the relevant quantity using the prescribed data, parameters, and equations without assuming the regularity conclusion. Equation (5.31) records such a proposed uniform estimate. Its use provides a direct way of expressing the claim over a class of initial fields.

The continuation proposition itself requires the finite bound (5.28). It does not require a closed-form expression for F_Ω. It also permits a proof that uses information about the solution, provided that information establishes the finite cumulative bound without assuming the conclusion being inferred. Thus an initial-data formula is a sufficient presentation of the control, rather than an exclusive form of admissible mathematical reasoning.

5.9.2 What solution-dependent and a posteriori mean here

The exact remainder g_{req} is evaluated from a solution's matrix balance. The standard and blockwise majorants also depend on that solution through its vorticity. Such dependence is compatible with (5.26). If a rigorous estimate controls the accumulated coefficient through a specified horizon, the proposition applies regardless of whether that control was obtained before or after a solution was calculated.

For a numerical trajectory, sampled values of the remainder and numerical quadrature give the displayed a posteriori envelope. Section 4.5 reports that numerical calculation. Its full-scalar residuals and time-step comparison document numerical agreement. They are not stated here as rigorous enclosures of the continuous PDE between samples or beyond the recorded interval. This distinction identifies the type of evidence already present; it does not discard the computed solution or change its observed result.

5.9.3 Why cumulative control is the condition

A smooth solution has finite spatial norms at each time strictly before T_*. This makes the coefficients finite on compact subintervals of its smooth interval. A continuation estimate additionally controls their accumulation as the interval approaches the proposed continuation time. That distinction is exactly the supremum in (5.28).

The velocity energy identity illustrates why the relevant estimates must be distinguished:

$$\nu \int_0^t Y(s)\, \mathrm{d}s \le \frac{1}{2} \|u_0\|_{L^2}^2 , \qquad \int_0^t g_{\mathrm{std}}[\omega](s)\, \mathrm{d}s = C_{\Omega,\delta} \nu^{-3} \int_0^t Y(s)^2\, \mathrm{d}s. \tag{5.33}$$

Control of the first integral is not a bound on the second integral: an L^1 bound on a nonnegative function does not alone control its L^2 norm. The blockwise estimate retains more information by replacing Y^2 with $\|P_\chi \omega\|_{L^2}^4$. This explains the mathematical content of the selected control quantity. It does not assert that any of the computed trajectories becomes singular.

For the global statement, a proof quantifies over the admissible class in one argument. It does not enumerate every possible initial field. Equation (5.31) is the paper's explicit universal claim, and the proposition proves its smooth-continuation consequence. The displayed majorant derivation and the completed numerical records do not by themselves establish that universal integral bound; its status in this paper is the theoretical assertion of the interpretation.

5.10 Reading the Completed Applications through the Same Estimate

Section 4 applies the same full representation to fixed amplitude-scaled initial fields and their recorded evolution. The exact construction $\mathcal{B}_\lambda = \lambda^3 \mathcal{B}_*$, $\mathcal{C}_\lambda = \lambda^2 \mathcal{C}_*$, and $\rho_\lambda = \lambda \rho_*$ explains the initial sign regimes. Those algebraic scaling laws concern the initial fields; the subsequent trajectories are computed under the same Navier–Stokes evolution.

The below case has zero recorded remainder and declining enstrophy. The boundary case starts with $\rho = 1$, so its remainder is positive because the retained margin uses $1 - \delta < 1$. The above case has a finite interval of stretching dominance before its recorded crossing into dissipation dominance. Its nonzero remainder represents the same surplus in (5.10). Table 4.3 and Figure 4.5 give the corresponding observed paths and envelopes.

The reported cumulative remainders are 0, 0.001349, and 0.022900 for the below, boundary, and above cases. Substituting them into $\exp(2A_g)$ gives the reported final normalized envelopes 1.000000, 1.002703, and 1.046865, respectively, to the stated precision. This connects the numerical display directly to (5.26). It does not substitute those values for the uncomputed analytic constants in $g_{\mathrm{blk},\chi}$.

The envelope follows from the same identity used to define its remainder. Its agreement with that shape is therefore a mathematical consequence of the construction. The application contributes the observed sizes, time histories, signs, and crossings, together with the independent full-matrix and enstrophy-rate consistency checks. These are the empirical contents of the display.

Transient enstrophy growth is compatible with the continuation estimate: the cumulative surplus may be positive and finite. Likewise, decay does not remove stretching, and growth does not remove viscosity. Both remain in the full Matrix of Error. The complex Fourier representation used by the solver encodes phases of the real physical field and leaves this real scalar balance unchanged. The

unforced system is the scope of the present estimate; physical forcing would contribute an additional term to its balance.

5.11 Result and Significance

The full visible chain is

$$\begin{gathered}\mathfrak{E}_{\mathrm{NS}} \longrightarrow \tfrac{1}{2}Y' = \mathcal{B} - \mathcal{C} \longrightarrow R_\delta \longrightarrow g_{\mathrm{req}} \le g_{\mathrm{blk},\chi}, \\ \sup_{t<\min\{T,T_*\}} \int_0^t g_{\mathrm{blk},\chi}(s)\,\mathrm{d}s < \infty \quad \Longrightarrow \quad \sup_{t<\min\{T,T_*\}} Y(t) < \infty \quad \Longrightarrow \quad T_* > T.\end{gathered} \tag{5.34}$$

The first line consists of the exact representation, definitions, and the derived blockwise bound. The second line is the proved conditional smooth-continuation implication. The assertion that its first condition holds for every admissible initial field and every finite horizon is the universal smooth-continuation interpretation proposed by the theory.

The completed results support that interpretation in specific ways: Section 2 supplies the full matrix and scalar identity; Section 3 validates their complete numerical realization and records a dissipative margin; Section 4 displays finite transient growth and zero and nonzero remainders under the same representation. The comparison with Chae–Choe and Miller locates the selected majorant in established regularity mathematics. The contribution advanced here is the connection of that structure to the cross-system Matrix-of-Error theory and its completed experimental realizations.

Nonlinear calculus was formed to evaluate complex real-world systems. Navier–Stokes is the most specific test in this program because its stretching and diffusion require the complete retained matrix geometry. The proposed universal solution representation and its smooth-continuation interpretation are stated together here, with their exact derivation, conditional consequence, and fixed empirical support visible. This is the paper's completed mathematical interpretation of the results.

Conclusion

The paper contains a three-part derivation-and-testing hierarchy, a fourth applications part, and a fifth part assembling their full Smooth-Continuation Interpretation. Section 1 first established the practical result: one shared dimension-agnostic derivation-and-testing process independently recovered known correction structures across Mercury, Burgers, and two- and three-dimensional Navier–Stokes. The systems supplied different variables and established term libraries, while the weak construction, joint identification, selection logic, controls, and reference checks remained common. This cross-system replication supports nonlinear calculus as a Universal Matrix-of-Error Theory over the completed range.

Section 2 then supplied the common mathematical form. Weakly evaluated term columns are weighted into a physical-contribution matrix; its correction submatrix records departures from the equilibrium case; its correction-column sum records their aggregate; and the full column sum records closure of the selected governing relation. Direct comparison with established matrix, sparse-identification, and weak-form methods shows where the individual operations already have independent support. Applied to three-dimensional vorticity, the shared map yields the complete Matrix of Error

$$\boxed{\mathfrak{E}_{\mathrm{NS}} = [S\omega \mid \nu\Delta\omega]},$$

with coefficient structure (B, b, s, C_2, C_1) under the exact two-plus-one decomposition.

Section 3 completed the predetermined test of that derivation. The expected full matrix closes its local stretching identity, spatial integral, diffusion partition, rotational transformation, and independent enstrophy-rate balance at approximately machine precision. The naive-form null fails in all 48 fixed flow/frame/split evaluations, and the matched 32^3 trajectory preserves the result. Across all 204 primary sampled flow-time states, viscous dissipation exceeds vortex stretching with $\mathcal{B}/\mathcal{C} \leq 0.955849$, and the sampled enstrophy values decrease along all four primary trajectories.

The completed sequence therefore reaches a positive and specific result: the full, coordinate-robust matrix derived in Section 2 is the validated three-dimensional Navier–Stokes realization of nonlinear calculus over the tested domain, while the naive form is insufficient. Its matrix coefficients retain every required stretching block together with viscous diffusion; its aggregate gives the observed enstrophy balance. The omni-dimensional-equilibrium hypothesis supplies the corresponding non-error case in which the Matrix of Error is zero.

Section 4 then used the unchanged matrix as an applications surface. Exact amplitude scaling placed one reference field at strict zero, dissipation dominance, nonzero aggregate balance, and stretching dominance. The dissipation-dominant path remained below the balance boundary and ended at 0.934690 of its initial enstrophy. The nonzero boundary path moved immediately into dissipation dominance and ended at 0.982692. The stretching-dominant path reached 1.034606, crossed the balance boundary at $t = 0.458898$, and ended at 1.034245. The complete five-block sum remained exact to approximately machine precision, and all three measured paths remained below their smooth-continuation envelopes. The resulting figures complete the previously proposed graphical application and show strict equilibrium, instantaneous nonzero balance, decay, finite transient growth, and a regime crossing as outputs of one representation.

From these findings, the Universal Matrix-of-Error Theory advances the full representation as the expected form for the Navier–Stokes existence-and-smoothness problem for all admissible smooth divergence-free initial fields and all finite times. The theory was formed to evaluate complex real-world systems; Navier–Stokes is the most specific physical test in this completed program.

Section 5 gives the full mathematical basis of the Smooth-Continuation Interpretation. The exact matrix balance yields the positive stretching surplus and its least nonnegative remainder g_{req}. The selected blockwise estimate proves $g_{\mathrm{req}} \le g_{\mathrm{blk},\chi}$, where $g_{\mathrm{blk},\chi} = C_{\Omega,\delta}\nu^{-3}\|P_\chi\omega\|_{L^2}^4$. Substitution into the enstrophy identity and integration give a definite conditional result: finite cumulative control of this majorant through a finite horizon guarantees smooth continuation through that horizon. The same conclusion follows from rigorous cumulative control of another majorant of g_{req}. Such control may use initial-data estimates or information about the solution itself.

The proposed universal smooth-continuation claim is that the selected blockwise majorant has finite cumulative integral on every finite horizon for every admissible initial field. Section 5 states that claim explicitly alongside its proved conditional implication and its relation to the established two-component regularity criteria. The exact representation and the reported numerical records support the interpretation advanced here; the universal integrability assertion is retained as a theoretical claim rather than reported as an additional computed or proved result.

The completed applications show how the interpretation reads physically: decreasing enstrophy, finite transient growth, and a balance crossing all occur within one retained matrix. Complex-valued spectral components encode phase and oscillation in the real physical field. External forcing or added physics would enter as further system-specific contributions. Together, the five parts present the cross-system support, full derivation, coordinate-robust validation, physical applications, and explicit Smooth-Continuation Interpretation of the Universal Matrix-of-Error Theory.

Preparation and Provenance Note

Christopher M Struck originated the nonlinear-calculus and omni-dimensional-equilibrium theories and proposed and directed the research program. He supplied the hypothesized integral-based framework, the Universal Matrix-of-Error Theory, the system-agnostic derivation-and-testing direction, the use of established systems as independent checks, and the initial (B, C_2, C_1) pre-derivation form. The full three-dimensional form was derived through the author-directed mathematical evaluation and then tested against that naive form as a fixed nested null. The author principally directed the interpretation of the mathematical and computational results. He selected the blockwise majorant used in Section 5.5 and directed the expansion of the smooth-continuation claim in Section 5. He has reviewed and takes responsibility for the claim advanced there.

Fable 5.1 began the computational research program and assisted with its initial adjustments, literature correspondences, experimental ladder, and initial three-dimensional implementation. GPT 5.6 drafted Section 1 from that work and assisted with the Experiment 8b continuation, literature and mathematical audits, formalization of the universal map and full-matrix comparison, implementation and audit of Experiments 9 and 10, preparation of the application figures, and assembly of the four-part working draft under the author's instructions and corrections. The first draft of Section 5, developed as an expansion of Section 3.7 through exchanges between the author and GPT 5.6, supplied substantial mathematical material retained in the final section.

GPT 6 assembled the final Section 5 at the author's request, consolidating and expanding that draft material together with the author-originated theory and jointly developed results of Sections 1–4 into the full Smooth-Continuation Interpretation. GPT 6 is specifically credited with the final assembly and detailed exposition of Section 5, including its visible derivation chain, the author-selected blockwise majorant, literature comparison, and consolidated statement of the proposed universal smooth-continuation claim. The earlier versions, four source drafts, and consolidated Parts I–III process artifact remain preserved separately.

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